



ITCS442

Relational Database Design

Project Phase 5: Physical Database Design and Tuning

Group 5: Restaurant Inventory Management System

Submitted by

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Presented to

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1. Database Application Requirements and Selected Transactions

The Restaurant Inventory Management System supports high-frequency operations such as order processing, payment handling, and inventory management. These operations generate large volumes of transactional data, especially in tables such as Orders, Payment, and Order_Menu, which contain tens of thousands of records.

To improve database performance, three performance-critical queries were selected based on their frequency of execution, data volume, and importance to business operations.

Data Entry Operations

- Enter the details for a new staff member.
- Enter the details for a new menu item (menu name, price, category, description).
- Enter the details for a new ingredient (ingredient name, unit, price per unit).
- Enter the details for a new supplier (supplier name, phone).
- Enter the details for a new customer order (order date, order status, total price).
- Enter payment details for an order (payment method, amount, paid at, payment status).
- Enter the details of menus included in an order (quantity, unit price at time).
- Enter the details for a new purchase order from a supplier (purchase date, status, total cost).
- Enter the details of ingredients included in a purchase order (quantity, unit cost).
- Enter inventory details when ingredients are stored (quantity available, last updated, minimum stock level).

Data Update / Deletion Operations

- Update or delete staff details.
- Update or delete menu details.
- Update or delete ingredient details.
- Update or delete supplier details.
- Update or delete order details.
- Update or delete order menu details (Order_Detail).
- Update or delete payment details.
- Update or delete purchase order details.
- Update or delete purchase order ingredient details (Purchase_Order_Detail).
- Update or delete inventory stock information.

Data Queries and Reports

- Retrieve orders processed by a specific staff member.
- List all customer orders including order date, status, and total price.
- Display payment details of a specific order.
- List menu items included in an order with quantity and unit price.
- Calculate the total price of an order based on order details.
- Identify top-selling menu items based on total quantity sold.
- Display ingredients required for a specific menu item.
- Display current inventory stock levels for all ingredients.
- Identify ingredients with stock below the minimum stock level.
- List suppliers and the ingredients they supply.
- Retrieve purchase orders from a specific supplier.
- Display ingredient details in a purchase order.
- Identify purchase orders with high unit cost items.

- Display purchase orders by status (Pending, Received, Cancelled).
- Identify inventory updates after receiving purchase orders.

Selected Transactions for Performance Optimization

The following four transactions are selected for performance optimization based on high execution frequency, large data volumes, and business criticality:

No.	Transaction Description	Reason for Selection
1	Update order status from Pending to Completed	High-frequency update on Orders table; executed on every order completion — uses Primary Key lookup for efficient row access
2	Identify top-selling menu items based on total quantity sold	Aggregation on large Order_Menu table (132,505 records); used for daily/weekly business reports
3	Identify ingredients with low stock level	Frequent daily-operations query on Inventory; critical for restocking decisions — trigger automatically alerts when low stock level.
4	Retrieve orders handled by a specific staff member	Frequent reporting query on Orders table (25,530 records); used for staff performance analysis and workload monitoring — filters by staff_id and orders results by date, benefiting from indexing to reduce full table scans

1. Update Order Status from Pending to Completed

Business Impact:

This operation is executed whenever a customer order is fulfilled and represents one of the most frequent write operations in the system. Any delay in updating the order status directly affects service speed and staff workflow.

Data Volume and Frequency:

The Orders table contains 25,530 records and continues to grow as new orders are placed. Status updates occur frequently during peak service periods, often dozens of times per hour, making this a high-frequency transaction.

Technical Reason:

Although the update targets a single row using the primary key (order_id), high concurrency conditions, where multiple staff update different orders simultaneously, can introduce write overhead. An index on order_status also supports queries that filter orders by status, such as retrieving all pending orders. The Payment table, which has a one-to-one relationship with Orders and contains the same number of records, may also be indirectly affected.

Why Selected:

Unlike bulk insert operations, this transaction is latency-sensitive. Any delay is immediately visible to staff during real-time service operations.

2. Identify Top-Selling Menu Items by Total Quantity Sold

Business Impact:

This report is used regularly by management to support decisions related to inventory planning, pricing, and promotions. Slow execution can delay decision-making processes.

Data Volume and Frequency:

The Order_Menu table is the largest table in the system, containing 132,505 records. This reflects approximately 4 to 5 menu items per order across 25,530 orders. The query performs aggregation using SUM and GROUP BY across the entire table and joins with the Menu table, which contains 75 records.

Technical Reason:

Without an index on Order_Menu.menu_id, the database must perform a full table scan before grouping and sorting the results. This makes it one of the most resource-intensive queries in the system. Adding an index on menu_id helps reduce join cost and improves overall performance during aggregation and sorting.

Why Selected:

This query operates on the largest dataset in the system and includes aggregation operations. Therefore, it shows a clear and measurable performance improvement after optimization.

3. Identify ingredients with low stock level

Business Impact:

This query is used by kitchen staff to monitor ingredient availability and ensure timely restocking. Failure to detect low stock levels may result in unavailable menu items and reduced customer satisfaction.

Data Volume and Frequency:

The Inventory table contains 120 records and the Ingredient table contains 125 records. Although relatively small, this query is executed multiple times per day. In addition, it is triggered automatically after inventory updates following purchase order processing, increasing its effective frequency.

Technical Reason:

The condition comparing quantity_available and minimum_stock_level requires a join between Inventory and Ingredient tables. Without indexing, the system performs a full scan and a nested loop join. Creating an index on inventory.ingredient_id improves join efficiency, while an index on quantity_available supports faster filtering and sorting.

Why Selected:

Despite smaller table sizes, the high frequency of execution combined with trigger-based invocation increases its overall impact on system performance. Optimization reduces repeated scanning and improves responsiveness.

4. Retrieve Orders Handled by a Specific Staff Member

Business Impact:

This query is used to monitor staff performance and workload by retrieving all orders handled by a specific staff member. It supports operational reporting and performance evaluation. Slow execution can affect reporting speed and decision-making during busy periods.

Data Volume and Frequency:

The Orders table contains 25,530 records and continues to grow over time. This query is executed frequently for reporting purposes, especially during peak hours when managers review staff performance. Each execution filters a subset of orders (e.g., 726 rows for a specific staff member).

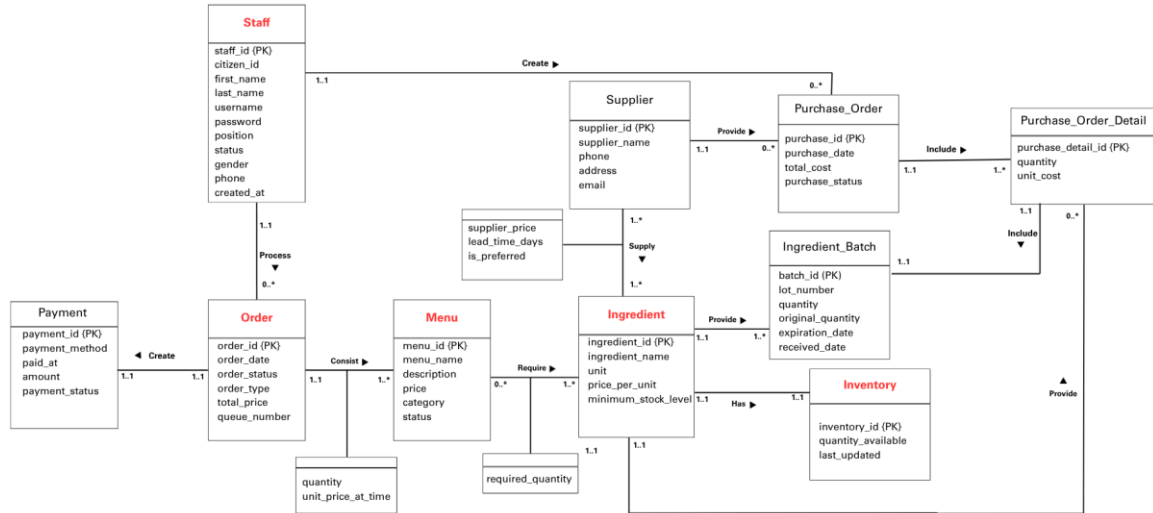
Technical Reason:

Without an index on `staff_id`, the database must perform a full Clustered Index Scan on the Orders table to locate matching records, resulting in unnecessary data access. By creating a non-clustered covering index on `staff_id` including `order_date`, `order_status`, `total_price`, and `order_type`, the optimizer can use an Index Seek to directly retrieve relevant rows. This significantly reduces I/O cost and improves query performance. The Staff table is accessed via a Clustered Index Seek using its primary key, which is already optimal.

Why Selected:

This query demonstrates a clear performance improvement from converting a full table scan into an index seek. It is also representative of real-world reporting queries that filter and sort data based on specific conditions, making it highly suitable for index optimization.

2. ER diagram



<https://canva.link/mggicon54x8hozv>

3.Relational Schema



<https://canva.link/elz5ijfj6z8k1ka>

4. The SQL commands only related to the selected transactions (or queries) and the specification of the related table, e.g., the number records in the table, the record sizes.

1. Update order status from Pending to Completed

	Queries	Specification of the related Table	Result																
	Data entry																		
A	Enter the details for a new customer order <pre>INSERT INTO Orders (order_id, order_date, order_status, order_type, total_price, queue_number, staff_id) VALUES ('ord26003', GETDATE(), 'Pending', 'Dine-in', 2000.55, '56', 10005);</pre>	Orders: 25530 records PK: order_id Operation: Insert	A new order record is created in the Orders table. This operation is performed whenever a customer places a new order. <table><tr><td></td><td>order_id</td><td>order_date</td><td>order_status</td><td>order_type</td><td>total_price</td><td>queue_number</td><td>staff_id</td></tr><tr><td>1</td><td>ord26003</td><td>2026-04-28 01:05:54.833</td><td>Pending</td><td>Dine-in</td><td>2000.55</td><td>56</td><td>10005</td></tr></table>		order_id	order_date	order_status	order_type	total_price	queue_number	staff_id	1	ord26003	2026-04-28 01:05:54.833	Pending	Dine-in	2000.55	56	10005
	order_id	order_date	order_status	order_type	total_price	queue_number	staff_id												
1	ord26003	2026-04-28 01:05:54.833	Pending	Dine-in	2000.55	56	10005												
B	Insert payment details for an order <pre>INSERT INTO Payment (payment_id, payment_method, amount, payment_status, paid_at, order_id) VALUES ('P26003', 'Cash', 2000.55, 'Paid', GETDATE(), 'ord26003');</pre>	Payment: 25530 records, PK: payment_id FK: order_id Operation: Insert	A new payment record is added for a specific order. This operation records transaction details after a payment is made. <table><tr><td></td><td>payment_id</td><td>payment_method</td><td>paid_at</td><td>amount</td><td>payment_status</td><td>order_id</td></tr><tr><td>1</td><td>P26003</td><td>Cash</td><td>2026-04-28 01:07:04.433</td><td>2000.55</td><td>Paid</td><td>ord26003</td></tr></table>		payment_id	payment_method	paid_at	amount	payment_status	order_id	1	P26003	Cash	2026-04-28 01:07:04.433	2000.55	Paid	ord26003		
	payment_id	payment_method	paid_at	amount	payment_status	order_id													
1	P26003	Cash	2026-04-28 01:07:04.433	2000.55	Paid	ord26003													
	Data Update																		

C	Update order details (Update Status) UPDATE Orders SET order_status = 'Completed' WHERE order_id = 'ord26003';	Orders: 25530 records PK: order_id Operation: Update (indexed attribute)	Updates the status of an existing order. Commonly used when the order process is completed or cancel. <table><tr><td></td><td>order_id</td><td>order_date</td><td>order_status</td><td>order_type</td><td>total_price</td><td>queue_number</td><td>staff_id</td></tr><tr><td>1</td><td>ord26003</td><td>2026-04-28 01:05:54.833</td><td>Completed</td><td>Dine-in</td><td>2000.55</td><td>56</td><td>10005</td></tr></table>		order_id	order_date	order_status	order_type	total_price	queue_number	staff_id	1	ord26003	2026-04-28 01:05:54.833	Completed	Dine-in	2000.55	56	10005																																																																																				
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	Data Queries																																																																																																						
D	List all customer orders including order date, status, and total price SELECT o order_id, o order_date, o order_status, o total_price FROM Orders o ORDER BY o order_date DESC; GO	Orders: 25,530 records PK: order_id Operation: Select Total rows: 25,530	Displays a list of all customer orders in the system, including order date, order status, and total price. This query is commonly used to monitor overall order activity and track the status of each order. The results are sorted by order date in descending order, showing the most recent orders first. <table><tr><td></td><td>order_id</td><td>order_date</td><td>order_status</td><td>total_price</td></tr><tr><td>1</td><td>ord22020</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>92.10</td></tr><tr><td>2</td><td>ord22021</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td></tr><tr><td>3</td><td>ord22022</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td></tr><tr><td>4</td><td>ord22023</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>505.04</td></tr><tr><td>5</td><td>ord22024</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>407.78</td></tr><tr><td>6</td><td>ord22025</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td></tr><tr><td>7</td><td>ord22026</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>73.62</td></tr><tr><td>8</td><td>ord22027</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>447.88</td></tr><tr><td>9</td><td>ord22028</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>118.56</td></tr><tr><td>10</td><td>ord22029</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>173.47</td></tr><tr><td>11</td><td>ord22030</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>108.39</td></tr><tr><td>12</td><td>ord22031</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>340.38</td></tr><tr><td>13</td><td>ord22032</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>292.11</td></tr><tr><td>14</td><td>ord22033</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>83.99</td></tr><tr><td>15</td><td>ord22034</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>252.28</td></tr><tr><td>16</td><td>ord22035</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>253.51</td></tr><tr><td>17</td><td>ord22036</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>501.46</td></tr><tr><td>18</td><td>ord22037</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>353.66</td></tr><tr><td>19</td><td>ord22038</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>296.85</td></tr></table>		order_id	order_date	order_status	total_price	1	ord22020	2026-03-31 23:59:00.000	Completed	92.10	2	ord22021	2026-03-31 23:59:00.000	Cancelled	0.00	3	ord22022	2026-03-31 23:59:00.000	Cancelled	0.00	4	ord22023	2026-03-31 23:59:00.000	Completed	505.04	5	ord22024	2026-03-31 23:59:00.000	Pending	407.78	6	ord22025	2026-03-31 23:59:00.000	Cancelled	0.00	7	ord22026	2026-03-31 23:59:00.000	Completed	73.62	8	ord22027	2026-03-31 23:59:00.000	Completed	447.88	9	ord22028	2026-03-31 23:59:00.000	Completed	118.56	10	ord22029	2026-03-31 23:59:00.000	Completed	173.47	11	ord22030	2026-03-31 23:59:00.000	Pending	108.39	12	ord22031	2026-03-31 23:59:00.000	Completed	340.38	13	ord22032	2026-03-31 23:59:00.000	Completed	292.11	14	ord22033	2026-03-31 23:59:00.000	Completed	83.99	15	ord22034	2026-03-31 23:59:00.000	Completed	252.28	16	ord22035	2026-03-31 23:59:00.000	Completed	253.51	17	ord22036	2026-03-31 23:59:00.000	Completed	501.46	18	ord22037	2026-03-31 23:59:00.000	Completed	353.66	19	ord22038	2026-03-31 23:59:00.000	Pending	296.85
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E	Display payment details of a specific order SELECT p payment_id, p payment_method, p amount, p payment_status, p paid_at FROM Payment p WHERE p order_id = 'ord26003'; GO	Payment: 25,530 records PK: payment_id FK: order_id Operation: Select (filtered) Total rows: 1	Displays the payment details for a specific order, including payment method, amount, payment status, and payment time. This query is used to verify whether a particular order has been successfully paid. It returns only one record when the order has a corresponding payment transaction. <table><tr><td></td><td>payment_id</td><td>payment_method</td><td>amount</td><td>payment_status</td><td>paid_at</td></tr><tr><td>1</td><td>P26003</td><td>Cash</td><td>2000.55</td><td>Paid</td><td>2026-04-28 22:57:26.440</td></tr></table>		payment_id	payment_method	amount	payment_status	paid_at	1	P26003	Cash	2000.55	Paid	2026-04-28 22:57:26.440																																																																																								
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2 Identify top-selling menu items by total quantity sold

	Queries	Specification of the related Table	Result														
	Data entry																
A	Enter the details for a new menu item (menu name, price, category, description). <pre>INSERT INTO Menu (menu_id, menu_name, description, price, category, status) VALUES ('mu0076', 'Pad Thai', 'Stir-fried rice noodles', 120.00, 'Main Course', 'Available'); GO</pre>	Menu: 75 records PK: menu_id Operation: Insert	A new menu item is created in the Menu table. This operation is performed when a new dish is added to the restaurant menu. <table><tr><th></th><th>menu_id</th><th>menu_name</th><th>description</th><th>price</th><th>category</th><th>status</th></tr><tr><td>1</td><td>mu0076</td><td>Pad Thai</td><td>Stir-fried rice noodles</td><td>120.00</td><td>Main Course</td><td>Available</td></tr></table>		menu_id	menu_name	description	price	category	status	1	mu0076	Pad Thai	Stir-fried rice noodles	120.00	Main Course	Available
	menu_id	menu_name	description	price	category	status											
1	mu0076	Pad Thai	Stir-fried rice noodles	120.00	Main Course	Available											
B	Enter the details for a new menu item <pre>INSERT INTO Order_Menu (order_id, menu_id, quantity, unit_price_at_time) VALUES ('ord26003', 'mu0076', 3, 120.00); GO</pre>	Order_Menu: 132,505 records PK: (order_id, menu_id) FK: order_id, menu_id Operation: Insert	A new record is inserted into the Order_Menu table. This operation records which menu items are included in a specific order, along with their quantity and unit price. <table><tr><th></th><th>order_id</th><th>menu_id</th><th>quantity</th><th>unit_price_at_time</th></tr><tr><td>1</td><td>ord26003</td><td>mu0076</td><td>3</td><td>120.00</td></tr></table>		order_id	menu_id	quantity	unit_price_at_time	1	ord26003	mu0076	3	120.00				
	order_id	menu_id	quantity	unit_price_at_time													
1	ord26003	mu0076	3	120.00													
	Data Update																
C	Update or delete order menu details (Order_Detail) <pre>--Update</pre>	Order_Menu: 132,505 records PK: (order_id, menu_id)	Update Updates the quantity or unit price of a menu item in an existing order. This operation is used when modifications are made to order details.														

	UPDATE Order_Menu SET quantity = 5, unit_price_at_time = 120.00 WHERE order_id = 'ord26003' AND menu_id = 'mu0076'; GO --Delete DELETE FROM Order_Menu WHERE order_id = 'ord26003' AND menu_id = 'mu0076'; GO	FK: order_id, menu_id Operation: Update / Delete	<table><tr><th></th><th>order_id</th><th>menu_id</th><th>quantity</th><th>unit_price_at_time</th></tr><tr><td>1</td><td>ord26003</td><td>mu0076</td><td>5</td><td>120.00</td></tr></table> Delete Removes a menu item from an order. This operation is performed when an item is cancelled or removed from the order. <table><tr><th></th><th>order_id</th><th>menu_id</th><th>quantity</th><th>unit_price_at_time</th></tr></table>		order_id	menu_id	quantity	unit_price_at_time	1	ord26003	mu0076	5	120.00		order_id	menu_id	quantity	unit_price_at_time																																								
	order_id	menu_id	quantity	unit_price_at_time																																																						
1	ord26003	mu0076	5	120.00																																																						
	order_id	menu_id	quantity	unit_price_at_time																																																						
	Data Queries																																																									
D	Identify top-selling menu items based on total quantity sold SELECT TOP 10 m.menu_id, m.menu_name, m.category, SUM(om.quantity) AS total_quantity_sold FROM Order_Menu om JOIN Menu m ON om.menu_id = m.menu_id GROUP BY m.menu_id, m.menu_name, m.category ORDER BY total_quantity_sold DESC;	Order_Menu: 132505records, FK: menu_id Menu: 75 records PK: menu_id Operation: Join + Group By + Aggregate (SUM) Total row : 10	Displays the top 10 best-selling menu items based on total quantity sold. Used for sales analysis and business insights. Results are sorted in descending order of total quantity sold. <table><tr><th></th><th>menu_id</th><th>menu_name</th><th>category</th><th>total_quantity_sold</th></tr><tr><td>1</td><td>mu0034</td><td>Chocolate Cake</td><td>Dessert</td><td>5610</td></tr><tr><td>2</td><td>mu0030</td><td>Orange Juice</td><td>Drink</td><td>5533</td></tr><tr><td>3</td><td>mu0055</td><td>Ice Cream Sundae</td><td>Dessert</td><td>5510</td></tr><tr><td>4</td><td>mu0028</td><td>Coke</td><td>Drink</td><td>5490</td></tr><tr><td>5</td><td>mu0013</td><td>Chicken Wings</td><td>Appetizer</td><td>5480</td></tr><tr><td>6</td><td>mu0022</td><td>Lemon Tea</td><td>Drink</td><td>5473</td></tr><tr><td>7</td><td>mu0019</td><td>Egg Fried Rice</td><td>Side</td><td>5449</td></tr><tr><td>8</td><td>mu0035</td><td>Cheesecake</td><td>Dessert</td><td>5441</td></tr><tr><td>9</td><td>mu0007</td><td>Pork Basil</td><td>Main Course</td><td>5430</td></tr><tr><td>10</td><td>mu0056</td><td>Grilled Chicken</td><td>Main Course</td><td>5423</td></tr></table>		menu_id	menu_name	category	total_quantity_sold	1	mu0034	Chocolate Cake	Dessert	5610	2	mu0030	Orange Juice	Drink	5533	3	mu0055	Ice Cream Sundae	Dessert	5510	4	mu0028	Coke	Drink	5490	5	mu0013	Chicken Wings	Appetizer	5480	6	mu0022	Lemon Tea	Drink	5473	7	mu0019	Egg Fried Rice	Side	5449	8	mu0035	Cheesecake	Dessert	5441	9	mu0007	Pork Basil	Main Course	5430	10	mu0056	Grilled Chicken	Main Course	5423
	menu_id	menu_name	category	total_quantity_sold																																																						
1	mu0034	Chocolate Cake	Dessert	5610																																																						
2	mu0030	Orange Juice	Drink	5533																																																						
3	mu0055	Ice Cream Sundae	Dessert	5510																																																						
4	mu0028	Coke	Drink	5490																																																						
5	mu0013	Chicken Wings	Appetizer	5480																																																						
6	mu0022	Lemon Tea	Drink	5473																																																						
7	mu0019	Egg Fried Rice	Side	5449																																																						
8	mu0035	Cheesecake	Dessert	5441																																																						
9	mu0007	Pork Basil	Main Course	5430																																																						
10	mu0056	Grilled Chicken	Main Course	5423																																																						

E	List menu items included in an order with quantity and unit price <pre>SELECT om.order_id , m.menu_name , om.quantity , om.unit_price_at_ time FROM Order_Menu om JOIN Menu m ON om.menu_id = m.menu_id; GO</pre>	Order_Menu: 132,505 records PK: (order_id, menu_id) FK: order_id, menu_id Menu: 75 records PK: menu_id Operation: Join + Select Total rows: 132,505	Displays all menu items included in each order, along with their quantity and unit price at the time of the order. This query is used to review order details and understand customer purchasing behavior. <table><tr><th></th><th>order_id</th><th>menu_name</th><th>quantity</th><th>unit_price_at_time</th></tr><tr><td>1</td><td>ord1</td><td>Green Curry</td><td>2</td><td>56.58</td></tr><tr><td>2</td><td>ord1</td><td>Tom Yum Soup</td><td>4</td><td>61.87</td></tr><tr><td>3</td><td>ord1</td><td>Chicken Basil</td><td>4</td><td>143.55</td></tr><tr><td>4</td><td>ord1</td><td>Fried Chicken</td><td>4</td><td>177.28</td></tr><tr><td>5</td><td>ord1</td><td>Grilled Pork</td><td>2</td><td>176.83</td></tr><tr><td>6</td><td>ord1</td><td>Beef Noodle</td><td>5</td><td>149.90</td></tr><tr><td>7</td><td>ord1</td><td>Spring Rolls</td><td>4</td><td>117.73</td></tr><tr><td>8</td><td>ord1</td><td>Chicken Wings</td><td>5</td><td>82.35</td></tr><tr><td>9</td><td>ord1</td><td>Steamed Rice</td><td>3</td><td>176.91</td></tr><tr><td>10</td><td>ord1</td><td>Garlic Rice</td><td>1</td><td>174.75</td></tr><tr><td>11</td><td>ord1</td><td>Cappuccino</td><td>5</td><td>90.40</td></tr><tr><td>12</td><td>ord1</td><td>Green Tea</td><td>1</td><td>79.72</td></tr><tr><td>13</td><td>ord1</td><td>Coke</td><td>3</td><td>97.68</td></tr><tr><td>14</td><td>ord1</td><td>Sprite</td><td>5</td><td>179.90</td></tr><tr><td>15</td><td>ord1</td><td>Coconut Ice Cream</td><td>5</td><td>45.37</td></tr><tr><td>16</td><td>ord1</td><td>Banana Fritter</td><td>2</td><td>32.51</td></tr><tr><td>17</td><td>ord1</td><td>Chocolate Cake</td><td>5</td><td>161.40</td></tr><tr><td>18</td><td>ord1</td><td>Chicken Noodle Soup</td><td>3</td><td>22.96</td></tr><tr><td>19</td><td>ord1</td><td>Fish Cake</td><td>4</td><td>53.25</td></tr><tr><td>20</td><td>ord1</td><td>Chicken Nuggets</td><td>5</td><td>192.84</td></tr><tr><td>21</td><td>ord1</td><td>Americano</td><td>2</td><td>98.61</td></tr><tr><td>22</td><td>ord1</td><td>Mocha</td><td>1</td><td>62.33</td></tr><tr><td>23</td><td>ord1</td><td>Matcha Latte</td><td>4</td><td>21.14</td></tr><tr><td>24</td><td>ord1</td><td>Pancake</td><td>5</td><td>190.39</td></tr><tr><td>25</td><td>ord1</td><td>Waffle</td><td>5</td><td>100.99</td></tr></table>		order_id	menu_name	quantity	unit_price_at_time	1	ord1	Green Curry	2	56.58	2	ord1	Tom Yum Soup	4	61.87	3	ord1	Chicken Basil	4	143.55	4	ord1	Fried Chicken	4	177.28	5	ord1	Grilled Pork	2	176.83	6	ord1	Beef Noodle	5	149.90	7	ord1	Spring Rolls	4	117.73	8	ord1	Chicken Wings	5	82.35	9	ord1	Steamed Rice	3	176.91	10	ord1	Garlic Rice	1	174.75	11	ord1	Cappuccino	5	90.40	12	ord1	Green Tea	1	79.72	13	ord1	Coke	3	97.68	14	ord1	Sprite	5	179.90	15	ord1	Coconut Ice Cream	5	45.37	16	ord1	Banana Fritter	2	32.51	17	ord1	Chocolate Cake	5	161.40	18	ord1	Chicken Noodle Soup	3	22.96	19	ord1	Fish Cake	4	53.25	20	ord1	Chicken Nuggets	5	192.84	21	ord1	Americano	2	98.61	22	ord1	Mocha	1	62.33	23	ord1	Matcha Latte	4	21.14	24	ord1	Pancake	5	190.39	25	ord1	Waffle	5	100.99
	order_id	menu_name	quantity	unit_price_at_time																																																																																																																																	
1	ord1	Green Curry	2	56.58																																																																																																																																	
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23	ord1	Matcha Latte	4	21.14																																																																																																																																	
24	ord1	Pancake	5	190.39																																																																																																																																	
25	ord1	Waffle	5	100.99																																																																																																																																	

F

Calculate the total price of an order based on order details

SELECT

om.order_id,

SUM(om.quantity

*

om.unit_price_at_

time) AS

total_price

FROM Order_Menu

om

GROUP BY

om.order_id;

GO

Order_Menu:

132,505

records

PK:

(order_id,

menu_id)

FK: order_id,

menu_id

Operation:

Group By +

Aggregate

Total rows:

25,530

(groups)

	order_id	total_price
1	ord1	14121.84
2	ord10	8304.03
3	ord100	9088.25
4	ord1000	8856.22
5	ord1001	12575.26
6	ord1002	9462.69
7	ord1003	12852.27
8	ord1004	11840.13
9	ord1005	12670.24
10	ord1006	9776.59
11	ord1007	13435.67
12	ord1008	9742.92
13	ord1009	10553.80
14	ord101	10753.31
15	ord1010	13002.46
16	ord1011	8365.33
17	ord1012	8945.65
18	ord1013	9161.65
19	ord1014	9195.68
20	ord1015	9601.39
21	ord1016	10106.67
22	ord1017	11288.93
23	ord1018	9489.81
24	ord1019	8842.02
25	ord102	8380.17

Calculates the total price of each order by summing the product of quantity and unit price for all menu items.

This query is used to verify the accuracy of order totals.

Each row represents one order with its calculated total price.

3 Identify ingredients with stock below minimum stock level

	Queries	Specification of the related Table	Result												
	Data entry														
A	Enter the details for a new ingredient <pre>INSERT INTO Ingredient (ingredient_id, ingredient_name, unit, price_per_unit, minimum_stock_level) VALUES ('ing0126', 'Basil', 'kg', 50.00, 5); GO</pre>	Ingredient: 125 records PK: ingredient_id Operation: Insert	A new ingredient is added to the Ingredient table. This operation defines the ingredient along with its unit, price, and minimum stock level required. <table><tr><td></td><td>ingredient_id</td><td>ingredient_name</td><td>unit</td><td>price_per_unit</td><td>minimum_stock_level</td></tr><tr><td>1</td><td>ing0126</td><td>Basil</td><td>kg</td><td>50.00</td><td>5</td></tr></table>		ingredient_id	ingredient_name	unit	price_per_unit	minimum_stock_level	1	ing0126	Basil	kg	50.00	5
	ingredient_id	ingredient_name	unit	price_per_unit	minimum_stock_level										
1	ing0126	Basil	kg	50.00	5										
B	Enter inventory details when ingredients are stored <pre>INSERT INTO Inventory (inventory_id, quantity_available, last_updated, ingredient_id) VALUES ('inv0121', 3, GETDATE(), 'ing0126'); GO</pre>	Inventory: 120 records PK: inventory_id FK: ingredient_id Operation: Insert	A new inventory record is created to track the available quantity of the ingredient. This operation is performed when ingredients are first stored in the inventory. <table><tr><td></td><td>inventory_id</td><td>quantity_available</td><td>last_updated</td><td>ingredient_id</td></tr><tr><td>1</td><td>inv0121</td><td>3</td><td>2026-04-29 00:52:22.157</td><td>ing0126</td></tr></table>		inventory_id	quantity_available	last_updated	ingredient_id	1	inv0121	3	2026-04-29 00:52:22.157	ing0126		
	inventory_id	quantity_available	last_updated	ingredient_id											
1	inv0121	3	2026-04-29 00:52:22.157	ing0126											
C	Enter the details of ingredients in a purchase order (restocking) <pre>INSERT INTO Purchase_Order_Detail (purchase_detail_id, quantity, unit_cost, purchase_id, ingredient_id) VALUES ('PD01751', 50,</pre>	Purchase_Order_Detail: 1,750 records PK: purchase_detail_id FK: purchase_id, ingredient_id Operation: Insert	A new purchase order detail record is created for restocking ingredients. This operation records the quantity and cost of ingredients ordered from suppliers. <table><tr><td></td><td>purchase_detail_id</td><td>quantity</td><td>unit_cost</td><td>purchase_id</td><td>ingredient_id</td></tr><tr><td>1</td><td>PD01751</td><td>50</td><td>50.00</td><td>PO0340</td><td>ing0126</td></tr></table>		purchase_detail_id	quantity	unit_cost	purchase_id	ingredient_id	1	PD01751	50	50.00	PO0340	ing0126
	purchase_detail_id	quantity	unit_cost	purchase_id	ingredient_id										
1	PD01751	50	50.00	PO0340	ing0126										

	<pre>50.00, 'P00340', 'ing0126'); GO</pre>																																																																																																																																																																																																																																																																		
	Data Update																																																																																																																																																																																																																																																																		
D	Update or delete inventory stock information <pre>UPDATE Inventory SET quantity_availabl e = quantity_availabl e - 26, last_updated =GETDATE() WHERE ingredient_id = 'ing0062'; GO</pre>	Inventory: 120 records FK: ingredient_id Operation: Update	Updates the available stock quantity of an ingredient in the inventory. This operation is commonly performed after ingredient usage or stock adjustment. <table><tr><td></td><td>inventory_id</td><td>quantity_available</td><td>last_updated</td><td>ingredient_id</td></tr><tr><td>1</td><td>inv0026</td><td>5</td><td>2026-04-29 01:24:49.233</td><td>ing0062</td></tr></table>		inventory_id	quantity_available	last_updated	ingredient_id	1	inv0026	5	2026-04-29 01:24:49.233	ing0062																																																																																																																																																																																																																																																						
	inventory_id	quantity_available	last_updated	ingredient_id																																																																																																																																																																																																																																																															
1	inv0026	5	2026-04-29 01:24:49.233	ing0062																																																																																																																																																																																																																																																															
E	Update purchase order ingredient details <pre>UPDATE Purchase_Order_De tail SET quantity = 60, unit_cost = 45.00 WHERE purchase_detail_i d = 'PD01751'; GO</pre>	Purchase_Or der_Detail: 1,750 records PK: purchase_det ail_id Operation: Update	Updates the quantity or unit cost of a specific ingredient in a purchase order. This operation is used when corrections or adjustments are required in procurement records. <table><tr><td></td><td>purchase_detail_id</td><td>quantity</td><td>unit_cost</td><td>purchase_id</td><td>ingredient_id</td></tr><tr><td>1</td><td>PD01751</td><td>60</td><td>45.00</td><td>P00340</td><td>ing0126</td></tr></table>		purchase_detail_id	quantity	unit_cost	purchase_id	ingredient_id	1	PD01751	60	45.00	P00340	ing0126																																																																																																																																																																																																																																																				
	purchase_detail_id	quantity	unit_cost	purchase_id	ingredient_id																																																																																																																																																																																																																																																														
1	PD01751	60	45.00	P00340	ing0126																																																																																																																																																																																																																																																														
	Data Queries																																																																																																																																																																																																																																																																		
F	Identify ingredients with stock below the minimum stock level <pre>SELECT i.ingredient_id, i.ingredient_name i.unit, i.minimum_stock_l evel, inv.quantity_avai lable, inv.last_updated, CASE WHEN inv.quantity_avai lable <</pre>	Inventory: 120 records FK: ingredient_id Ingredient: 125 records PK: ingredient_id Operation: Join + Filtering + Sorting Total row : 120	Displays ingredients with stock levels below the minimum threshold. Used to support restocking decisions in daily operations. Results are sorted by the lowest available quantity first. <table><tr><td></td><td>ingredient_id</td><td>ingredient_name</td><td>unit</td><td>minimum_stock_level</td><td>quantity_available</td><td>last_updated</td><td>stock_status</td></tr><tr><td>1</td><td>ing0126</td><td>Basil</td><td>kg</td><td>5</td><td>3</td><td>2026-04-29 00:52:22.157</td><td>LOW</td></tr><tr><td>2</td><td>ing0062</td><td>Ginger</td><td>kg</td><td>5</td><td>5</td><td>2026-04-29 01:24:49.233</td><td>SUFFICIENT</td></tr><tr><td>3</td><td>ing0093</td><td>Yeast</td><td>kg</td><td>5</td><td>22</td><td>2026-03-07 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>4</td><td>ing0020</td><td>Carrot</td><td>kg</td><td>5</td><td>25</td><td>2026-03-03 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>5</td><td>ing0062</td><td>Ginger</td><td>kg</td><td>5</td><td>26</td><td>2026-04-29 01:24:49.233</td><td>SUFFICIENT</td></tr><tr><td>6</td><td>ing0092</td><td>Baking Soda</td><td>kg</td><td>5</td><td>26</td><td>2026-03-03 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>7</td><td>ing0108</td><td>Sunflower Seed</td><td>kg</td><td>5</td><td>28</td><td>2026-03-09 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>8</td><td>ing0091</td><td>Baking Powder</td><td>kg</td><td>5</td><td>31</td><td>2026-03-06 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>9</td><td>ing0116</td><td>Glass Noodles</td><td>kg</td><td>5</td><td>31</td><td>2026-04-04 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>10</td><td>ing0006</td><td>Beef</td><td>kg</td><td>5</td><td>32</td><td>2026-03-05 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>11</td><td>ing0109</td><td>Pumpkin Seed</td><td>kg</td><td>5</td><td>33</td><td>2026-03-05 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>12</td><td>ing0089</td><td>Wheat Flour</td><td>kg</td><td>10</td><td>34</td><td>2026-03-02 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>13</td><td>ing0068</td><td>Bean Sprout</td><td>kg</td><td>5</td><td>34</td><td>2026-03-04 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>14</td><td>ing0094</td><td>Vanilla Extract</td><td>liter</td><td>5</td><td>36</td><td>2026-03-01 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>15</td><td>ing0115</td><td>Egg Noodles</td><td>kg</td><td>5</td><td>36</td><td>2026-03-08 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>16</td><td>ing0076</td><td>Cardamom</td><td>kg</td><td>5</td><td>37</td><td>2026-03-07 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>17</td><td>ing0018</td><td>Tomato</td><td>kg</td><td>5</td><td>38</td><td>2026-03-07 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>18</td><td>ing0110</td><td>Chia Seed</td><td>kg</td><td>5</td><td>38</td><td>2026-03-02 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>19</td><td>ing0119</td><td>Dried Banana</td><td>kg</td><td>5</td><td>39</td><td>2026-03-10 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>20</td><td>ing0098</td><td>Condensed Coconut Milk</td><td>can</td><td>10</td><td>41</td><td>2026-03-02 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>21</td><td>ing0120</td><td>Coconut Milk</td><td>liter</td><td>10</td><td>42</td><td>2026-03-02 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>22</td><td>ing0114</td><td>Rice Noodles</td><td>kg</td><td>5</td><td>42</td><td>2026-03-01 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>23</td><td>ing0088</td><td>Corn Flour</td><td>kg</td><td>5</td><td>42</td><td>2026-03-09 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>24</td><td>ing0113</td><td>Cornflakes</td><td>kg</td><td>5</td><td>43</td><td>2026-03-01 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>25</td><td>ing0030</td><td>Potato</td><td>kg</td><td>5</td><td>44</td><td>2026-03-02 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>26</td><td>ing0034</td><td>Butter</td><td>kg</td><td>5</td><td>46</td><td>2026-03-10 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>27</td><td>ing0116</td><td>Glass Noodles</td><td>kg</td><td>5</td><td>46</td><td>2026-03-08 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>28</td><td>ing0108</td><td>Sunflower Seed</td><td>kg</td><td>5</td><td>46</td><td>2026-03-01 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>29</td><td>ing0113</td><td>Cornflakes</td><td>kg</td><td>5</td><td>47</td><td>2026-03-06 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>30</td><td>ing0083</td><td>White Sugar</td><td>kg</td><td>10</td><td>47</td><td>2026-03-06 00:00:00.000</td><td>SUFFICIENT</td></tr><tr><td>31</td><td>ing0064</td><td>Mint</td><td>kg</td><td>5</td><td>47</td><td>2026-03-03 00:00:00.000</td><td>SUFFICIENT</td></tr></table>		ingredient_id	ingredient_name	unit	minimum_stock_level	quantity_available	last_updated	stock_status	1	ing0126	Basil	kg	5	3	2026-04-29 00:52:22.157	LOW	2	ing0062	Ginger	kg	5	5	2026-04-29 01:24:49.233	SUFFICIENT	3	ing0093	Yeast	kg	5	22	2026-03-07 00:00:00.000	SUFFICIENT	4	ing0020	Carrot	kg	5	25	2026-03-03 00:00:00.000	SUFFICIENT	5	ing0062	Ginger	kg	5	26	2026-04-29 01:24:49.233	SUFFICIENT	6	ing0092	Baking Soda	kg	5	26	2026-03-03 00:00:00.000	SUFFICIENT	7	ing0108	Sunflower Seed	kg	5	28	2026-03-09 00:00:00.000	SUFFICIENT	8	ing0091	Baking Powder	kg	5	31	2026-03-06 00:00:00.000	SUFFICIENT	9	ing0116	Glass Noodles	kg	5	31	2026-04-04 00:00:00.000	SUFFICIENT	10	ing0006	Beef	kg	5	32	2026-03-05 00:00:00.000	SUFFICIENT	11	ing0109	Pumpkin Seed	kg	5	33	2026-03-05 00:00:00.000	SUFFICIENT	12	ing0089	Wheat Flour	kg	10	34	2026-03-02 00:00:00.000	SUFFICIENT	13	ing0068	Bean Sprout	kg	5	34	2026-03-04 00:00:00.000	SUFFICIENT	14	ing0094	Vanilla Extract	liter	5	36	2026-03-01 00:00:00.000	SUFFICIENT	15	ing0115	Egg Noodles	kg	5	36	2026-03-08 00:00:00.000	SUFFICIENT	16	ing0076	Cardamom	kg	5	37	2026-03-07 00:00:00.000	SUFFICIENT	17	ing0018	Tomato	kg	5	38	2026-03-07 00:00:00.000	SUFFICIENT	18	ing0110	Chia Seed	kg	5	38	2026-03-02 00:00:00.000	SUFFICIENT	19	ing0119	Dried Banana	kg	5	39	2026-03-10 00:00:00.000	SUFFICIENT	20	ing0098	Condensed Coconut Milk	can	10	41	2026-03-02 00:00:00.000	SUFFICIENT	21	ing0120	Coconut Milk	liter	10	42	2026-03-02 00:00:00.000	SUFFICIENT	22	ing0114	Rice Noodles	kg	5	42	2026-03-01 00:00:00.000	SUFFICIENT	23	ing0088	Corn Flour	kg	5	42	2026-03-09 00:00:00.000	SUFFICIENT	24	ing0113	Cornflakes	kg	5	43	2026-03-01 00:00:00.000	SUFFICIENT	25	ing0030	Potato	kg	5	44	2026-03-02 00:00:00.000	SUFFICIENT	26	ing0034	Butter	kg	5	46	2026-03-10 00:00:00.000	SUFFICIENT	27	ing0116	Glass Noodles	kg	5	46	2026-03-08 00:00:00.000	SUFFICIENT	28	ing0108	Sunflower Seed	kg	5	46	2026-03-01 00:00:00.000	SUFFICIENT	29	ing0113	Cornflakes	kg	5	47	2026-03-06 00:00:00.000	SUFFICIENT	30	ing0083	White Sugar	kg	10	47	2026-03-06 00:00:00.000	SUFFICIENT	31	ing0064	Mint	kg	5	47	2026-03-03 00:00:00.000	SUFFICIENT
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31	ing0064	Mint	kg	5	47	2026-03-03 00:00:00.000	SUFFICIENT																																																																																																																																																																																																																																																												

	<pre>i.minimum_stock_level THEN 'LOW' ELSE 'SUFFICIENT' END AS stock_status FROM Inventory inv JOIN Ingredient i ON inv.ingredient_id = i.ingredient_id ORDER BY inv.quantity_available ASC;</pre>																																																																																																																																																																																																		
G	Display current inventory stock levels for all ingredients <pre>SELECT i.ingredient_id, i.ingredient_name , iunit, inv.quantity_available, inv.last_updated FROM Inventory inv JOIN Ingredient i ON inv.ingredient_id = i.ingredient_id ORDER BY inv.quantity_available ASC; GO</pre>	Inventory: 120 records FK: ingredient_id Ingredient: 125 records PK: ingredient_id Operation: Join + Select Total rows: 120	Displays the current stock levels of all ingredients in the inventory. This query provides an overview of inventory <table><tr><th></th><th>ingredient_id</th><th>ingredient_name</th><th>unit</th><th>quantity_available</th><th>last_updated</th></tr><tr><td>1</td><td>ing0126</td><td>Basil</td><td>kg</td><td>3</td><td>2026-04-29 00:52:22.157</td></tr><tr><td>2</td><td>ing0062</td><td>Ginger</td><td>kg</td><td>5</td><td>2026-04-29 01:24:49.233</td></tr><tr><td>3</td><td>ing0093</td><td>Yeast</td><td>kg</td><td>22</td><td>2026-03-07 00:00:00.000</td></tr><tr><td>4</td><td>ing0020</td><td>Carrot</td><td>kg</td><td>25</td><td>2026-03-03 00:00:00.000</td></tr><tr><td>5</td><td>ing0062</td><td>Ginger</td><td>kg</td><td>26</td><td>2026-04-29 01:24:49.233</td></tr><tr><td>6</td><td>ing0092</td><td>Baking Soda</td><td>kg</td><td>26</td><td>2026-03-03 00:00:00.000</td></tr><tr><td>7</td><td>ing0108</td><td>Sunflower Seed</td><td>kg</td><td>28</td><td>2026-03-09 00:00:00.000</td></tr><tr><td>8</td><td>ing0091</td><td>Baking Powder</td><td>kg</td><td>31</td><td>2026-03-06 00:00:00.000</td></tr><tr><td>9</td><td>ing0116</td><td>Glass Noodles</td><td>kg</td><td>31</td><td>2026-04-04 00:00:00.000</td></tr><tr><td>10</td><td>ing0006</td><td>Beef</td><td>kg</td><td>32</td><td>2026-03-05 00:00:00.000</td></tr><tr><td>11</td><td>ing0109</td><td>Pumpkin Seed</td><td>kg</td><td>33</td><td>2026-03-05 00:00:00.000</td></tr><tr><td>12</td><td>ing0089</td><td>Wheat Flour</td><td>kg</td><td>34</td><td>2026-03-02 00:00:00.000</td></tr><tr><td>13</td><td>ing0068</td><td>Bean Sprout</td><td>kg</td><td>34</td><td>2026-03-04 00:00:00.000</td></tr><tr><td>14</td><td>ing0094</td><td>Vanilla Extract</td><td>liter</td><td>36</td><td>2026-03-01 00:00:00.000</td></tr><tr><td>15</td><td>ing0115</td><td>Egg Noodles</td><td>kg</td><td>36</td><td>2026-03-08 00:00:00.000</td></tr><tr><td>16</td><td>ing0076</td><td>Cardamom</td><td>kg</td><td>37</td><td>2026-03-07 00:00:00.000</td></tr><tr><td>17</td><td>ing0018</td><td>Tomato</td><td>kg</td><td>38</td><td>2026-03-07 00:00:00.000</td></tr><tr><td>18</td><td>ing0110</td><td>Chia Seed</td><td>kg</td><td>38</td><td>2026-03-02 00:00:00.000</td></tr><tr><td>19</td><td>ing0119</td><td>Dried Banana</td><td>kg</td><td>39</td><td>2026-03-10 00:00:00.000</td></tr><tr><td>20</td><td>ing0098</td><td>Condensed Coconut Milk</td><td>can</td><td>41</td><td>2026-03-02 00:00:00.000</td></tr><tr><td>21</td><td>ing0120</td><td>Coconut Milk</td><td>liter</td><td>42</td><td>2026-03-02 00:00:00.000</td></tr><tr><td>22</td><td>ing0114</td><td>Rice Noodles</td><td>kg</td><td>42</td><td>2026-03-01 00:00:00.000</td></tr><tr><td>23</td><td>ing0088</td><td>Corn Flour</td><td>kg</td><td>42</td><td>2026-03-09 00:00:00.000</td></tr><tr><td>24</td><td>ing0113</td><td>Cornflakes</td><td>kg</td><td>43</td><td>2026-03-01 00:00:00.000</td></tr><tr><td>25</td><td>ing0030</td><td>Potato</td><td>kg</td><td>44</td><td>2026-03-02 00:00:00.000</td></tr><tr><td>26</td><td>ing0034</td><td>Butter</td><td>kg</td><td>46</td><td>2026-03-10 00:00:00.000</td></tr><tr><td>27</td><td>ing0116</td><td>Glass Noodles</td><td>kg</td><td>46</td><td>2026-03-08 00:00:00.000</td></tr><tr><td>28</td><td>ing0108</td><td>Sunflower Seed</td><td>kg</td><td>46</td><td>2026-03-01 00:00:00.000</td></tr><tr><td>29</td><td>ing0113</td><td>Cornflakes</td><td>kg</td><td>47</td><td>2026-03-06 00:00:00.000</td></tr><tr><td>30</td><td>ing0083</td><td>White Sugar</td><td>kg</td><td>47</td><td>2026-03-06 00:00:00.000</td></tr><tr><td>31</td><td>ing0064</td><td>Mint</td><td>kg</td><td>47</td><td>2026-03-03 00:00:00.000</td></tr></table> status for daily monitoring.		ingredient_id	ingredient_name	unit	quantity_available	last_updated	1	ing0126	Basil	kg	3	2026-04-29 00:52:22.157	2	ing0062	Ginger	kg	5	2026-04-29 01:24:49.233	3	ing0093	Yeast	kg	22	2026-03-07 00:00:00.000	4	ing0020	Carrot	kg	25	2026-03-03 00:00:00.000	5	ing0062	Ginger	kg	26	2026-04-29 01:24:49.233	6	ing0092	Baking Soda	kg	26	2026-03-03 00:00:00.000	7	ing0108	Sunflower Seed	kg	28	2026-03-09 00:00:00.000	8	ing0091	Baking Powder	kg	31	2026-03-06 00:00:00.000	9	ing0116	Glass Noodles	kg	31	2026-04-04 00:00:00.000	10	ing0006	Beef	kg	32	2026-03-05 00:00:00.000	11	ing0109	Pumpkin Seed	kg	33	2026-03-05 00:00:00.000	12	ing0089	Wheat Flour	kg	34	2026-03-02 00:00:00.000	13	ing0068	Bean Sprout	kg	34	2026-03-04 00:00:00.000	14	ing0094	Vanilla Extract	liter	36	2026-03-01 00:00:00.000	15	ing0115	Egg Noodles	kg	36	2026-03-08 00:00:00.000	16	ing0076	Cardamom	kg	37	2026-03-07 00:00:00.000	17	ing0018	Tomato	kg	38	2026-03-07 00:00:00.000	18	ing0110	Chia Seed	kg	38	2026-03-02 00:00:00.000	19	ing0119	Dried Banana	kg	39	2026-03-10 00:00:00.000	20	ing0098	Condensed Coconut Milk	can	41	2026-03-02 00:00:00.000	21	ing0120	Coconut Milk	liter	42	2026-03-02 00:00:00.000	22	ing0114	Rice Noodles	kg	42	2026-03-01 00:00:00.000	23	ing0088	Corn Flour	kg	42	2026-03-09 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29	ing0113	Cornflakes	kg	47	2026-03-06 00:00:00.000																																																																																																																																																																																														
30	ing0083	White Sugar	kg	47	2026-03-06 00:00:00.000																																																																																																																																																																																														
31	ing0064	Mint	kg	47	2026-03-03 00:00:00.000																																																																																																																																																																																														
H	Identify inventory updates after receiving purchase orders <pre>SELECT inv.inventory_id, i.ingredient_name , inv.quantity_available, inv.last_updated FROM Inventory inv</pre>	Inventory: 120 records FK: ingredient_id Ingredient: 125 records PK: ingredient_id Operation: Join + Filter by date	Displays recently updated inventory records based on the last updated timestamp. This query is used to monitor inventory changes after restocking activities. Results are sorted by the most recent updates first.																																																																																																																																																																																																

	<pre>JOIN Ingredient i ON inv.ingredient_id = i.ingredient_id WHERE inv.last_updated >= DATEADD(DAY, -1, GETDATE()) ORDER BY inv.last_updated DESC; GO</pre>	Total rows: 6	<table><tr><th></th><th>inventory_id</th><th>ingredient_name</th><th>quantity_available</th><th>last_updated</th></tr><tr><td>1</td><td>inv0119</td><td>Dried Banana</td><td>66</td><td>2026-05-05 00:00:00.000</td></tr><tr><td>2</td><td>inv0120</td><td>Coconut Milk</td><td>71</td><td>2026-05-03 00:00:00.000</td></tr><tr><td>3</td><td>inv0118</td><td>Dried Mango</td><td>62</td><td>2026-05-02 00:00:00.000</td></tr><tr><td>4</td><td>inv0026</td><td>Ginger</td><td>5</td><td>2026-04-29 01:24:49.233</td></tr><tr><td>5</td><td>inv0062</td><td>Ginger</td><td>26</td><td>2026-04-29 01:24:49.233</td></tr><tr><td>6</td><td>inv0121</td><td>Basil</td><td>3</td><td>2026-04-29 00:52:22.157</td></tr></table>		inventory_id	ingredient_name	quantity_available	last_updated	1	inv0119	Dried Banana	66	2026-05-05 00:00:00.000	2	inv0120	Coconut Milk	71	2026-05-03 00:00:00.000	3	inv0118	Dried Mango	62	2026-05-02 00:00:00.000	4	inv0026	Ginger	5	2026-04-29 01:24:49.233	5	inv0062	Ginger	26	2026-04-29 01:24:49.233	6	inv0121	Basil	3	2026-04-29 00:52:22.157																																																																																																																																																																																													
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6	inv0121	Basil	3	2026-04-29 00:52:22.157																																																																																																																																																																																																																															
I	<pre>List suppliers and the ingredients they supply SELECT s.supplier_id, s.supplier_name, i.ingredient_name , si.supplier_price , s.lead_time_days , si.is_preferred FROM Supplier_Ingrdie nt si JOIN Supplier s ON si.supplier_id = s.supplier_id JOIN Ingredient i ON si.ingredient_id = i.ingredient_id ORDER BY s.supplier_name; GO</pre>	Supplier: ~50 records PK: supplier_id Supplier_Ingr edient: ~800 records FK: supplier_id, ingredient_id Ingredient: 125 records PK: ingredient_id Operation: Join (3 tables) Total rows: 800	Displays the relationship between suppliers and the ingredients they provide, including pricing and lead time information. This query supports supplier selection and restocking decisions. <table><tr><th></th><th>supplier_id</th><th>supplier_name</th><th>ingredient_name</th><th>supplier_price</th><th>lead_time_days</th><th>is_preferred</th></tr><tr><td>1</td><td>sup1015</td><td>Asian Import Export</td><td>Rice</td><td>256.12</td><td>4</td><td>0</td></tr><tr><td>2</td><td>sup1015</td><td>Asian Import Export</td><td>Pork</td><td>195.33</td><td>2</td><td>1</td></tr><tr><td>3</td><td>sup1015</td><td>Asian Import Export</td><td>Shrimp</td><td>36.34</td><td>2</td><td>0</td></tr><tr><td>4</td><td>sup1015</td><td>Asian Import Export</td><td>Squid</td><td>178.78</td><td>3</td><td>0</td></tr><tr><td>5</td><td>sup1015</td><td>Asian Import Export</td><td>Onion</td><td>297.89</td><td>3</td><td>1</td></tr><tr><td>6</td><td>sup1015</td><td>Asian Import Export</td><td>Lime</td><td>42.04</td><td>5</td><td>1</td></tr><tr><td>7</td><td>sup1015</td><td>Asian Import Export</td><td>Carrot</td><td>294.72</td><td>3</td><td>1</td></tr><tr><td>8</td><td>sup1015</td><td>Asian Import Export</td><td>Cabbage</td><td>271.80</td><td>5</td><td>1</td></tr><tr><td>9</td><td>sup1015</td><td>Asian Import Export</td><td>Mushroom</td><td>198.95</td><td>5</td><td>1</td></tr><tr><td>10</td><td>sup1015</td><td>Asian Import Export</td><td>Spinach</td><td>150.95</td><td>2</td><td>0</td></tr><tr><td>11</td><td>sup1015</td><td>Asian Import Export</td><td>Potato</td><td>285.35</td><td>4</td><td>1</td></tr><tr><td>12</td><td>sup1015</td><td>Asian Import Export</td><td>Milk</td><td>104.78</td><td>2</td><td>0</td></tr><tr><td>13</td><td>sup1015</td><td>Asian Import Export</td><td>Cheese</td><td>298.22</td><td>3</td><td>1</td></tr><tr><td>14</td><td>sup1015</td><td>Asian Import Export</td><td>Sugar</td><td>179.04</td><td>5</td><td>0</td></tr><tr><td>15</td><td>sup1015</td><td>Asian Import Export</td><td>Fish Sauce</td><td>172.66</td><td>3</td><td>1</td></tr><tr><td>16</td><td>sup1015</td><td>Asian Import Export</td><td>Vinegar</td><td>198.98</td><td>4</td><td>1</td></tr><tr><td>17</td><td>sup1015</td><td>Asian Import Export</td><td>Red Onion</td><td>163.35</td><td>2</td><td>0</td></tr><tr><td>18</td><td>sup1015</td><td>Asian Import Export</td><td>Kaffir Lime Leaf</td><td>209.46</td><td>5</td><td>0</td></tr><tr><td>19</td><td>sup1015</td><td>Asian Import Export</td><td>Turmeric</td><td>234.57</td><td>1</td><td>0</td></tr><tr><td>20</td><td>sup1015</td><td>Asian Import Export</td><td>White Pepper</td><td>243.14</td><td>4</td><td>0</td></tr><tr><td>21</td><td>sup1015</td><td>Asian Import Export</td><td>Fennel Seed</td><td>160.08</td><td>5</td><td>1</td></tr><tr><td>22</td><td>sup1015</td><td>Asian Import Export</td><td>Tamarind Paste</td><td>106.65</td><td>4</td><td>1</td></tr><tr><td>23</td><td>sup1015</td><td>Asian Import Export</td><td>Coconut Sugar</td><td>153.65</td><td>2</td><td>0</td></tr><tr><td>24</td><td>sup1015</td><td>Asian Import Export</td><td>Brown Sugar</td><td>262.66</td><td>2</td><td>0</td></tr><tr><td>25</td><td>sup1015</td><td>Asian Import Export</td><td>White Sugar</td><td>95.74</td><td>4</td><td>0</td></tr><tr><td>26</td><td>sup1015</td><td>Asian Import Export</td><td>Maple Syrup</td><td>191.84</td><td>3</td><td>0</td></tr><tr><td>27</td><td>sup1015</td><td>Asian Import Export</td><td>Corn Flour</td><td>229.15</td><td>4</td><td>0</td></tr><tr><td>28</td><td>sup1015</td><td>Asian Import Export</td><td>Vanilla Extract</td><td>69.75</td><td>2</td><td>0</td></tr><tr><td>29</td><td>sup1015</td><td>Asian Import Export</td><td>Dark Chocolate</td><td>280.20</td><td>4</td><td>1</td></tr><tr><td>30</td><td>sup1015</td><td>Asian Import Export</td><td>White Chocolate</td><td>180.56</td><td>2</td><td>0</td></tr><tr><td>31</td><td>sup1015</td><td>Asian Import Export</td><td>Condensed Co...</td><td>75.01</td><td>5</td><td>0</td></tr></table>		supplier_id	supplier_name	ingredient_name	supplier_price	lead_time_days	is_preferred	1	sup1015	Asian Import Export	Rice	256.12	4	0	2	sup1015	Asian Import Export	Pork	195.33	2	1	3	sup1015	Asian Import Export	Shrimp	36.34	2	0	4	sup1015	Asian Import Export	Squid	178.78	3	0	5	sup1015	Asian Import Export	Onion	297.89	3	1	6	sup1015	Asian Import Export	Lime	42.04	5	1	7	sup1015	Asian Import Export	Carrot	294.72	3	1	8	sup1015	Asian Import Export	Cabbage	271.80	5	1	9	sup1015	Asian Import Export	Mushroom	198.95	5	1	10	sup1015	Asian Import Export	Spinach	150.95	2	0	11	sup1015	Asian Import Export	Potato	285.35	4	1	12	sup1015	Asian Import Export	Milk	104.78	2	0	13	sup1015	Asian Import Export	Cheese	298.22	3	1	14	sup1015	Asian Import Export	Sugar	179.04	5	0	15	sup1015	Asian Import Export	Fish Sauce	172.66	3	1	16	sup1015	Asian Import Export	Vinegar	198.98	4	1	17	sup1015	Asian Import Export	Red Onion	163.35	2	0	18	sup1015	Asian Import Export	Kaffir Lime Leaf	209.46	5	0	19	sup1015	Asian Import Export	Turmeric	234.57	1	0	20	sup1015	Asian Import Export	White Pepper	243.14	4	0	21	sup1015	Asian Import Export	Fennel Seed	160.08	5	1	22	sup1015	Asian Import Export	Tamarind Paste	106.65	4	1	23	sup1015	Asian Import Export	Coconut Sugar	153.65	2	0	24	sup1015	Asian Import Export	Brown Sugar	262.66	2	0	25	sup1015	Asian Import Export	White Sugar	95.74	4	0	26	sup1015	Asian Import Export	Maple Syrup	191.84	3	0	27	sup1015	Asian Import Export	Corn Flour	229.15	4	0	28	sup1015	Asian Import Export	Vanilla Extract	69.75	2	0	29	sup1015	Asian Import Export	Dark Chocolate	280.20	4	1	30	sup1015	Asian Import Export	White Chocolate	180.56	2	0	31	sup1015	Asian Import Export	Condensed Co...	75.01	5	0
	supplier_id	supplier_name	ingredient_name	supplier_price	lead_time_days	is_preferred																																																																																																																																																																																																																													
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4 .Retrieve orders processed by a specific staff member

	Queries	Specification of the related Table	Result																								
	Data entry																										
A	Enter the details for a new staff member <pre>INSERT INTO Staff (staff_id, citizen_id, first_name, last_name, username, password, position, status, gender, phone, created_at) VALUES (10041, '1234567890123', 'John', 'Doe', 'johndoe', 'password123', 'Waiter', 'Active', 'Male', '0812345678', GETDATE()); GO</pre>	Staff: 40 records PK: staff_id Operation: Insert	A new staff record is created in the Staff table. This operation is performed when a new employee joins the restaurant. <table><tr><th></th><th>staff_id</th><th>citizen_id</th><th>first_name</th><th>last_name</th><th>username</th><th>password</th><th>position</th><th>status</th><th>gender</th><th>phone</th><th>created_at</th></tr><tr><td>1</td><td>10041</td><td>1234567890123</td><td>John</td><td>Doe</td><td>johndoe</td><td>password123</td><td>Waiter</td><td>Active</td><td>Male</td><td>0812345678</td><td>2026-04-29 01:41:24.273</td></tr></table>		staff_id	citizen_id	first_name	last_name	username	password	position	status	gender	phone	created_at	1	10041	1234567890123	John	Doe	johndoe	password123	Waiter	Active	Male	0812345678	2026-04-29 01:41:24.273
	staff_id	citizen_id	first_name	last_name	username	password	position	status	gender	phone	created_at																
1	10041	1234567890123	John	Doe	johndoe	password123	Waiter	Active	Male	0812345678	2026-04-29 01:41:24.273																
B	Enter the details for a new customer order <pre>INSERT INTO Orders (order_id, order_date, order_status, order_type, total_price, queue_number, staff_id) VALUES ('ord26003', GETDATE(), 'Pending', 'Dine-in', 2000.55, '56', 10005);</pre>	Orders: 25530 records PK: order_id Operation: Insert	A new order record is created in the Orders table. This operation is performed whenever a customer places a new order. <table><tr><th></th><th>order_id</th><th>order_date</th><th>order_status</th><th>order_type</th><th>total_price</th><th>queue_number</th><th>staff_id</th></tr><tr><td>1</td><td>ord26003</td><td>2026-04-28 01:05:54.833</td><td>Pending</td><td>Dine-in</td><td>2000.55</td><td>56</td><td>10005</td></tr></table>		order_id	order_date	order_status	order_type	total_price	queue_number	staff_id	1	ord26003	2026-04-28 01:05:54.833	Pending	Dine-in	2000.55	56	10005								
	order_id	order_date	order_status	order_type	total_price	queue_number	staff_id																				
1	ord26003	2026-04-28 01:05:54.833	Pending	Dine-in	2000.55	56	10005																				

	Data Update																																																																																																																																																																																																																																																																		
C	Update order details (change staff or status) UPDATE Orders SET order_status = 'Completed' WHERE order_id = 'ord26003';	Orders: 25530 records PK: order_id Operation: Update (indexed attribute)	Updates the status of an existing order. Commonly used when the order process is completed or cancel. <table><tr><td></td><td>order_id</td><td>order_date</td><td>order_status</td><td>order_type</td><td>total_price</td><td>queue_number</td><td>staff_id</td></tr><tr><td>1</td><td>ord26003</td><td>2026-04-28 01:05:54.833</td><td>Completed</td><td>Dine-in</td><td>2000.55</td><td>56</td><td>10005</td></tr></table>		order_id	order_date	order_status	order_type	total_price	queue_number	staff_id	1	ord26003	2026-04-28 01:05:54.833	Completed	Dine-in	2000.55	56	10005																																																																																																																																																																																																																																																
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	Data Queries																																																																																																																																																																																																																																																																		
D	Retrieve orders processed by a specific staff member (T4) SELECT o.order_id, o.order_date, o.order_status, o.total_price, o.order_type, s.first_name, s.last_name FROM Orders o JOIN Staff s ON o.staff_id = s.staff_id WHERE o.staff_id = 10006 ORDER BY o.order_date DESC; GO	Orders: 25,530 records PK: order_id FK: staff_id Staff: 40 records PK: staff_id Operation: Join + Filter by staff_id Total rows: 726	Displays all orders handled by a specific staff member, including order date, status, and total price. This query is used to track staff performance and workload. Results are sorted by order date in descending order. <table><tr><td></td><td>order_id</td><td>order_date</td><td>order_status</td><td>total_price</td><td>order_type</td><td>first_name</td><td>last_name</td></tr><tr><td>1</td><td>ord22046</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>91.73</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>2</td><td>ord22053</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>353.96</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>3</td><td>ord22068</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>223.26</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>4</td><td>ord22091</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>504.55</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>5</td><td>ord22099</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>111.21</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr><tr><td>6</td><td>ord22156</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>256.65</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>7</td><td>ord22243</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>160.74</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>8</td><td>ord22247</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>9</td><td>ord22337</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>494.78</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>10</td><td>ord22346</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>499.78</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>11</td><td>ord22351</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>244.28</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>12</td><td>ord22519</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>104.63</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr><tr><td>13</td><td>ord22597</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>406.13</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>14</td><td>ord22711</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>475.44</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr><tr><td>15</td><td>ord22740</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>440.19</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>16</td><td>ord22744</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>17</td><td>ord22769</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>398.67</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>18</td><td>ord22774</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>192.81</td><td>Dine-in</td><td>Sophia</td><td>Taylor</td></tr><tr><td>19</td><td>ord22892</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>482.97</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr><tr><td>20</td><td>ord22913</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>459.83</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr><tr><td>21</td><td>ord22923</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>267.99</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr><tr><td>22</td><td>ord22939</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>315.31</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr><tr><td>23</td><td>ord23127</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>349.48</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>24</td><td>ord23143</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>241.63</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>25</td><td>ord23146</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>397.30</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>26</td><td>ord23164</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>257.57</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>27</td><td>ord23215</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>390.90</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>28</td><td>ord23218</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>221.29</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>29</td><td>ord23232</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>481.86</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>30</td><td>ord23243</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td><td>Delivery</td><td>Sophia</td><td>Taylor</td></tr><tr><td>31</td><td>ord23254</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>61.84</td><td>Takeaway</td><td>Sophia</td><td>Taylor</td></tr></table>		order_id	order_date	order_status	total_price	order_type	first_name	last_name	1	ord22046	2026-03-31 23:59:00.000	Completed	91.73	Dine-in	Sophia	Taylor	2	ord22053	2026-03-31 23:59:00.000	Completed	353.96	Dine-in	Sophia	Taylor	3	ord22068	2026-03-31 23:59:00.000	Completed	223.26	Delivery	Sophia	Taylor	4	ord22091	2026-03-31 23:59:00.000	Completed	504.55	Dine-in	Sophia	Taylor	5	ord22099	2026-03-31 23:59:00.000	Completed	111.21	Takeaway	Sophia	Taylor	6	ord22156	2026-03-31 23:59:00.000	Completed	256.65	Dine-in	Sophia	Taylor	7	ord22243	2026-03-31 23:59:00.000	Completed	160.74	Dine-in	Sophia	Taylor	8	ord22247	2026-03-31 23:59:00.000	Cancelled	0.00	Delivery	Sophia	Taylor	9	ord22337	2026-03-31 23:59:00.000	Completed	494.78	Dine-in	Sophia	Taylor	10	ord22346	2026-03-31 23:59:00.000	Pending	499.78	Dine-in	Sophia	Taylor	11	ord22351	2026-03-31 23:59:00.000	Completed	244.28	Dine-in	Sophia	Taylor	12	ord22519	2026-03-31 23:59:00.000	Completed	104.63	Takeaway	Sophia	Taylor	13	ord22597	2026-03-31 23:59:00.000	Completed	406.13	Dine-in	Sophia	Taylor	14	ord22711	2026-03-31 23:59:00.000	Pending	475.44	Takeaway	Sophia	Taylor	15	ord22740	2026-03-31 23:59:00.000	Completed	440.19	Delivery	Sophia	Taylor	16	ord22744	2026-03-31 23:59:00.000	Cancelled	0.00	Delivery	Sophia	Taylor	17	ord22769	2026-03-31 23:59:00.000	Pending	398.67	Delivery	Sophia	Taylor	18	ord22774	2026-03-31 23:59:00.000	Completed	192.81	Dine-in	Sophia	Taylor	19	ord22892	2026-03-31 23:59:00.000	Completed	482.97	Takeaway	Sophia	Taylor	20	ord22913	2026-03-31 23:59:00.000	Completed	459.83	Takeaway	Sophia	Taylor	21	ord22923	2026-03-31 23:59:00.000	Completed	267.99	Takeaway	Sophia	Taylor	22	ord22939	2026-03-31 23:59:00.000	Completed	315.31	Takeaway	Sophia	Taylor	23	ord23127	2026-03-31 23:59:00.000	Completed	349.48	Delivery	Sophia	Taylor	24	ord23143	2026-03-31 23:59:00.000	Completed	241.63	Delivery	Sophia	Taylor	25	ord23146	2026-03-31 23:59:00.000	Completed	397.30	Delivery	Sophia	Taylor	26	ord23164	2026-03-31 23:59:00.000	Completed	257.57	Delivery	Sophia	Taylor	27	ord23215	2026-03-31 23:59:00.000	Completed	390.90	Delivery	Sophia	Taylor	28	ord23218	2026-03-31 23:59:00.000	Completed	221.29	Delivery	Sophia	Taylor	29	ord23232	2026-03-31 23:59:00.000	Pending	481.86	Delivery	Sophia	Taylor	30	ord23243	2026-03-31 23:59:00.000	Cancelled	0.00	Delivery	Sophia	Taylor	31	ord23254	2026-03-31 23:59:00.000	Pending	61.84	Takeaway	Sophia	Taylor
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31	ord23254	2026-03-31 23:59:00.000	Pending	61.84	Takeaway	Sophia	Taylor																																																																																																																																																																																																																																																												
E	List all customer orders SELECT o.order_id, o.order_date, o.order_status, o.total_price FROM Orders o	Orders: 25,530 records PK: order_id Operation: Select Total rows: 25,530	Displays all customer orders in the system, including order date, status, and total price. This query is used for general monitoring of order activity. Results are sorted by order date in descending order.																																																																																																																																																																																																																																																																

	ORDER BY o order_date DESC; GO		<table><tr><th></th><th>order_id</th><th>order_date</th><th>order_status</th><th>total_price</th></tr><tr><td>1</td><td>ord26003</td><td>2026-04-28 22:57:26.440</td><td>Pending</td><td>2000.55</td></tr><tr><td>2</td><td>ord22020</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>92.10</td></tr><tr><td>3</td><td>ord22021</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td></tr><tr><td>4</td><td>ord22022</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td></tr><tr><td>5</td><td>ord22023</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>505.04</td></tr><tr><td>6</td><td>ord22024</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>407.78</td></tr><tr><td>7</td><td>ord22025</td><td>2026-03-31 23:59:00.000</td><td>Cancelled</td><td>0.00</td></tr><tr><td>8</td><td>ord22026</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>73.62</td></tr><tr><td>9</td><td>ord22027</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>447.88</td></tr><tr><td>10</td><td>ord22028</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>118.56</td></tr><tr><td>11</td><td>ord22029</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>173.47</td></tr><tr><td>12</td><td>ord22030</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>108.39</td></tr><tr><td>13</td><td>ord22031</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>340.38</td></tr><tr><td>14</td><td>ord22032</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>292.11</td></tr><tr><td>15</td><td>ord22033</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>83.99</td></tr><tr><td>16</td><td>ord22034</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>252.28</td></tr><tr><td>17</td><td>ord22035</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>253.51</td></tr><tr><td>18</td><td>ord22036</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>501.46</td></tr><tr><td>19</td><td>ord22037</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>353.66</td></tr><tr><td>20</td><td>ord22038</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>296.85</td></tr><tr><td>21</td><td>ord22039</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>132.55</td></tr><tr><td>22</td><td>ord22040</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>211.57</td></tr><tr><td>23</td><td>ord22041</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>460.74</td></tr><tr><td>24</td><td>ord22042</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>328.94</td></tr><tr><td>25</td><td>ord22043</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>521.30</td></tr><tr><td>26</td><td>ord22044</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>514.36</td></tr><tr><td>27</td><td>ord22045</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>342.35</td></tr><tr><td>28</td><td>ord22046</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>91.73</td></tr><tr><td>29</td><td>ord22047</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>437.27</td></tr><tr><td>30</td><td>ord22048</td><td>2026-03-31 23:59:00.000</td><td>Pending</td><td>493.58</td></tr><tr><td>31</td><td>ord22049</td><td>2026-03-31 23:59:00.000</td><td>Completed</td><td>398.18</td></tr></table>		order_id	order_date	order_status	total_price	1	ord26003	2026-04-28 22:57:26.440	Pending	2000.55	2	ord22020	2026-03-31 23:59:00.000	Completed	92.10	3	ord22021	2026-03-31 23:59:00.000	Cancelled	0.00	4	ord22022	2026-03-31 23:59:00.000	Cancelled	0.00	5	ord22023	2026-03-31 23:59:00.000	Completed	505.04	6	ord22024	2026-03-31 23:59:00.000	Pending	407.78	7	ord22025	2026-03-31 23:59:00.000	Cancelled	0.00	8	ord22026	2026-03-31 23:59:00.000	Completed	73.62	9	ord22027	2026-03-31 23:59:00.000	Completed	447.88	10	ord22028	2026-03-31 23:59:00.000	Completed	118.56	11	ord22029	2026-03-31 23:59:00.000	Completed	173.47	12	ord22030	2026-03-31 23:59:00.000	Pending	108.39	13	ord22031	2026-03-31 23:59:00.000	Completed	340.38	14	ord22032	2026-03-31 23:59:00.000	Completed	292.11	15	ord22033	2026-03-31 23:59:00.000	Completed	83.99	16	ord22034	2026-03-31 23:59:00.000	Completed	252.28	17	ord22035	2026-03-31 23:59:00.000	Completed	253.51	18	ord22036	2026-03-31 23:59:00.000	Completed	501.46	19	ord22037	2026-03-31 23:59:00.000	Completed	353.66	20	ord22038	2026-03-31 23:59:00.000	Pending	296.85	21	ord22039	2026-03-31 23:59:00.000	Completed	132.55	22	ord22040	2026-03-31 23:59:00.000	Completed	211.57	23	ord22041	2026-03-31 23:59:00.000	Completed	460.74	24	ord22042	2026-03-31 23:59:00.000	Completed	328.94	25	ord22043	2026-03-31 23:59:00.000	Completed	521.30	26	ord22044	2026-03-31 23:59:00.000	Pending	514.36	27	ord22045	2026-03-31 23:59:00.000	Completed	342.35	28	ord22046	2026-03-31 23:59:00.000	Completed	91.73	29	ord22047	2026-03-31 23:59:00.000	Completed	437.27	30	ord22048	2026-03-31 23:59:00.000	Pending	493.58	31	ord22049	2026-03-31 23:59:00.000	Completed	398.18
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F	Display payment details of orders handled by the staff SELECT p.payment_id, p.payment_method, p.amount, p.payment_status, p.paid_at, o.order_id FROM Payment p JOIN Orders o ON p.order_id = o.order_id WHERE o.staff_id = 10002;	Payment: 25,530 records PK: payment_id FK: order_id Orders: 25,530 records FK: staff_id Operation: Join + Filter by staff_id Total rows: 657	Displays payment details for orders processed by a specific staff member. This includes payment method, amount, payment status, and payment time. This query is used to verify transactions associated with each staff member.																																																																																																																																																																

	payment_id	payment_method	amount	payment_status	paid_at	order_id
1	P10006	QR Payment	368.89	Paid	2026-03-15 07:32:00.000	ord10006
2	P10021	QR Payment	254.91	Paid	2026-03-15 08:02:00.000	ord10021
3	P10038	Cash	95.37	Pending	2026-03-24 11:30:56.000	ord10038
4	P10176	Credit Card	0.00	Paid	2026-03-15 13:12:00.000	ord10176
5	P1019	QR Payment	335.87	Paid	2026-03-02 19:58:00.000	ord01019
6	P10191	QR Payment	426.89	Paid	2026-03-15 13:42:00.000	ord10191
7	P10202	QR Payment	307.73	Paid	2026-03-15 14:04:00.000	ord10202
8	P10308	Credit Card	0.00	Paid	2026-03-15 17:36:00.000	ord10308
9	P10359	Cash	0.00	Paid	2026-03-15 19:18:00.000	ord10359
10	P10370	QR Payment	179.14	Pending	2026-03-24 11:30:56.000	ord10370
11	P10377	QR Payment	457.30	Paid	2026-03-15 19:54:00.000	ord10377
12	P10382	Credit Card	143.06	Pending	2026-03-24 11:30:56.000	ord10382
13	P10392	Credit Card	312.98	Pending	2026-03-24 11:30:56.000	ord10392
14	P10409	QR Payment	285.67	Paid	2026-03-15 20:58:00.000	ord10409
15	P10458	Cash	0.00	Paid	2026-03-15 22:36:00.000	ord10458
16	P10460	QR Payment	315.74	Paid	2026-03-15 22:40:00.000	ord10460
17	P10484	Credit Card	94.29	Paid	2026-03-15 23:28:00.000	ord10484
18	P10512	Credit Card	0.00	Failed	2026-03-16 00:24:00.000	ord10512
19	P10526	Cash	307.40	Pending	2026-03-24 11:30:56.000	ord10526
20	P1058	Cash	142.91	Paid	2026-03-02 21:16:00.000	ord01058
21	P10598	QR Payment	202.53	Pending	2026-03-24 11:30:56.000	ord10598
22	P10629	Cash	144.55	Paid	2026-03-16 04:18:00.000	ord10629
23	P10676	Credit Card	235.92	Paid	2026-03-16 05:52:00.000	ord10676
24	P10685	Credit Card	0.00	Failed	2026-03-16 06:10:00.000	ord10685
25	P1071	QR Payment	168.47	Paid	2026-03-02 21:42:00.000	ord01071
26	P10733	QR Payment	245.10	Paid	2026-03-16 07:46:00.000	ord10733
27	P10783	Cash	523.63	Pending	2026-03-24 11:30:56.000	ord10783
28	P10788	Credit Card	80.01	Paid	2026-03-16 09:36:00.000	ord10788
29	P10804	Cash	416.18	Pending	2026-03-24 11:30:56.000	ord10804
30	P10808	Cash	212.18	Pending	2026-03-24 11:30:56.000	ord10808
31	P10820	QR Payment	355.95	Paid	2026-03-16 10:40:00.000	ord10820

5. The results of the analysis of each selected transaction before improvement. The result from RDBMS's query analyzer can also be attached (as a figure).

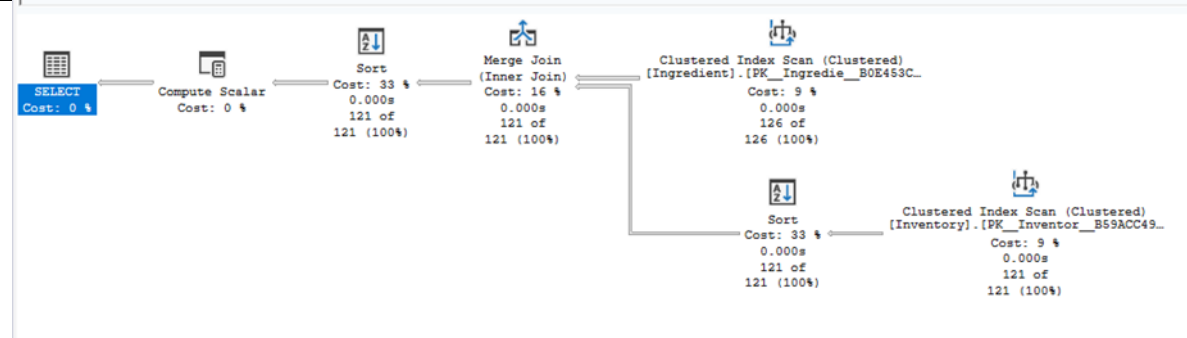
1. Update order status from Pending to Completed

Transaction Analysis Form					
			April 27, 2026		
Transaction	Update order status from Pending to Completed				
Volume	Average:	50 per hour			
	Peak:	200 per hour (during 12:00-14:00 and 18:00-21:00)			
UPDATE Orders SET order_status = 'Completed' WHERE order_id = 'ord26003'; GO		Predicate:	order_id = 'ord26003' (WHERE on Primary Key)		
		Join attributes:	-		
		Ordering attribute:	-		
		Group attribute:	-		
		Built-in functions:	-		
		Attributes updated:	order_status		
T-SQL UPDATE Cost: 0 % Clustered Index Update [Orders].[PK__Orders__46596229236AC... Cost: 100 % 0.000s 1 of 1 (100%)					
Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour
1	Orders	R	1	1	200
2	Orders	U	1	1	200
Total References			2	2	400

2. Identify top-selling menu items based on total quantity sold

Transaction Analysis Form					
					April 27, 2026
Transaction		Identify top-selling menu items based on total quantity sold			
Volume		Average:	5 per hour		
		Peak:	20 per hour (during report generation periods)		
<pre>SELECT TOP 10 m.menu_id, m.menu_name, m.category, SUM(om.quantity) AS total_quantity_sold FROM Order_Menu om JOIN Menu m ON om.menu_id = m.menu_id GROUP BY m.menu_id, m.menu_name, m.category ORDER BY total_quantity_sold DESC; GO</pre>			Predicate:	-	
			Join attributes:	Order_Menu.menu_id = Menu.menu_id	
			Ordering attribute:	total_quantity_sold DESC	
			Group attribute:	menu_id, menu_name, category	
			Built-in functions:	SUM(quantity)	
			Attributes updated:	-	
Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour
1	Order_Menu	R	132,505	132,505	2,650,100
2	Menu	R	10	10	1500
Total References			132,515	132,515	2,651,600

3. Identify ingredients with low stock level

Transaction Analysis Form					
					April 27, 2026
Transaction	Identify ingredients with low stock level				
Volume	Average:	10 per hour			
	Peak:	30 per hour (during kitchen prep hours 08:00-10:00)			
<pre>SELECT i.ingredient_id, i.ingredient_name, i.unit, i.minimum_stock_level, inv.quantity_available, inv.last_updated, CASE WHEN inv.quantity_available < i.minimum_stock_level THEN 'LOW' ELSE 'SUFFICIENT' END AS stock_status FROM Inventory inv JOIN Ingredient i ON invingredient_id = iingredient_id ORDER BY inv.quantity_available ASC; GO</pre>			Predicate:	-	
			Join attributes:	Inventory.ingredient_id = Ingredient.ingredient_id	
			Ordering attribute:	inv.quantity_available ASC;	
			Group attribute:	-	
			Built-in functions:	CASE WHEN inv.quantity_available < i.minimum_stock_level THEN 'LOW' ELSE 'SUFFICIENT' END	
			Attributes updated:	-	
					
Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour
1	Inventory	R	120	1,210	3,600
2	Ingredient	R	121	1,260	3,630
Total References			241	2,470	7,320

4. Identify orders handled by a specific staff member

Transaction Analysis Form					
					April 27, 2026
Transaction	Identify orders handled by a specific staff member				
Volume	Average:	5 per hour			
	Peak:	20 per hour (during report generation periods)			
SELECT o order_id, o order_date, o order_status, o total_price, o order_type, s first_name, s last_name FROM Orders o JOIN Staff s ON ostaff_id = sstaff_id WHERE ostaff_id = 10006 ORDER BY o.order_date DESC; GO			Predicate:	o.staff_id = 10006	
			Join attributes:	Orders.staff_id = Staff.staff_id	
			Ordering attribute:	o.order_date DESC	
			Group attribute:	-	
			Built-in functions:	-	
			Attributes updated:	-	
Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour
1	Orders	R	726	3,630	14,520
2	Staff	R	1	5	20
Total			727	3,635	14,540

6. The results of the analysis of each selected transaction after improvement. The result from RDBMS's query analyzer can also be attached (as a figure).

1. Update order status from Pending to Completed(After: Parameterized Query)

Transaction Analysis Form					
					April 27, 2026
Transaction		Update order status from Pending to Completed (After: Parameterized Query)			
Volume		Average:	50 per hour		
		Peak:	200 per hour (during 12:00-14:00 and 18:00-21:00)		
<pre>CREATE PROCEDURE UpdateOrderStatus @order_id NVARCHAR(10), @order_status NVARCHAR(20) AS BEGIN SET NOCOUNT ON; UPDATE Orders SET order_status = @order_status WHERE order_id = @order_id; END; GO CREATE INDEX idx_orders_status ON Orders(order_status); GO EXEC UpdateOrderStatus @order_id = 'ord26003', @order_status = 'Completed'; GO</pre>			Predicate:	order_id = @order_id (parameterized, eliminates implicit conversion)	
			Join attributes:	-	
			Ordering attribute:	-	
			Group attribute:	-	
			Built-in functions:	-	
			Attributes updated:	order_status	

T-SQL

UPDATE
Cost: 0 %

Clustered Index Update
Cost: 86 %
0.000s
1 of
1 (100%)

Compute Scalar
Cost: 0 %
0.000s
1 of
1 (100%)

Compute Scalar
Cost: 0 %

Nested Loops (Inner Join)
Cost: 0 %
0.000s
1 of
1 (100%)

Compute Scalar
Cost: 0 %
0.000s
1 of
1 (100%)

Constant Scan
Cost: 0 %
0.000s
1 of
1 (100%)

Clustered Index Seek (Clustered)
[Orders].[PK_Orders_46596229236AC...]
Cost: 14 %
0.000s
1 of
1 (100%)

Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour
1	Orders	R	1	1	200
2	Orders	U	1	1	200
Total References			2	2	400

2. Identify top-selling menu items based on total quantity sold (After: Covering Index)

Transaction Analysis Form					
					April 27, 2026
Transaction	Identify top-selling menu items based on total quantity sold (After: Covering Index)				
Volume	Average:	5 per hour			
	Peak:	20 per hour (during report generation periods)			
<pre>CREATE PROCEDURE GetTopSellingMenus @TopN INT = 10 AS BEGIN SET NOCOUNT ON; SELECT TOP (@TopN) m.menu_id, m.menu_name, m.category, SUM(om.quantity) AS total_quantity_sold FROM Order_Menu om JOIN Menu m ON om.menu_id = m.menu_id GROUP BY m.menu_id, m.menu_name, m.category ORDER BY total_quantity_sold DESC; END; GO CREATE INDEX idx_ordermenu_menuid_qty ON Order_Menu(menu_id) INCLUDE (quantity); GO EXEC GetTopSellingMenus @TopN = 10; GO</pre>			Predicate:	-	
			Join attributes:	Order_Menu.menu_id = Menu.menu_id	
			Ordering attribute:	total_quantity_sold DESC	
			Group attribute:	menu_id, menu_name, category	
			Built-in functions:	SUM(quantity)	
Attributes updated:	-				

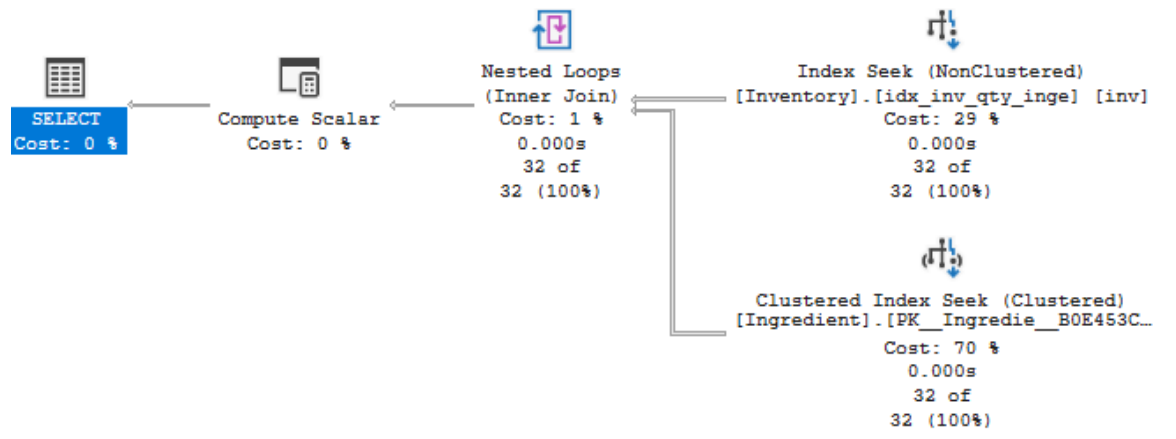
The diagram illustrates the execution plan for the provided query. It starts with an 'Index Scan (NonClustered)' on the [Order_Menu].[idx_ordermenu_menuid_...] index, which is the most expensive operation at 84% cost. This is followed by a 'Stream Aggregate (Aggregate)' operation at 13% cost. The data then flows through a 'Sort' operation (2% cost) and a 'Nested Loops (Inner Join)' operation (0% cost). Finally, a 'Top' operation (0% cost) returns the results. A 'Clustered Index Seek (Clustered)' on the [Menu].[PK_Menu_4CA0FADC4CF0C91A]... index is also shown, which is used by the 'Nested Loops' operation. The plan includes various statistics such as cost, time, and row counts for each step.

Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour
1	Order_Menu	R	132,505	132,505	2,650,100
2	Menu	R	10	10	1500
Total References			132,515	132,515	2,651,600

3. Identify ingredients with low stock level (After: Non-Clustered Index)

Filtering (quantity_available < 50) increases selectivity, reducing I/O and allowing index-based access.

Transaction Analysis Form					
April 27, 2026					
Transaction	Identify ingredients with low stock level				
Volume	Average:	10 per hour			
	Peak:	30 per hour (during kitchen prep hours 08:00-10:00)			
<pre>CREATE INDEX idx_inv_qty_inge ON Inventory(quantity_available, ingredient_id) INCLUDE (last_updated); GO SELECT iingredient_id, iingredient_name, iunit, iminimum_stock_level, invquantity_available, invlast_updated, 'LOW' AS stock_status FROM Inventory inv JOIN Ingredient i ON invingredient_id = iingredient_id WHERE invquantity_available < 50 ORDER BY inv.quantity_available ASC; GO</pre>			Predicate:	inv.quantity_available < 50	
			Join attributes:	Inventory.ingredient_id = Ingredient.ingredient_id	
			Ordering attribute:	inv.quantity_available ASC	
			Group attribute:	-	
			Built-in functions:	'LOW' AS stock_status	
			Attributes updated:	-	



```
graph LR
    SELECT[SELECT  
Cost: 0 %] --> CS[Compute Scalar  
Cost: 0 %]
    CS --> NL[Nested Loops  
(Inner Join)  
Cost: 1 %  
0.000s  
32 of  
32 (100%)]
    NL --> IS[Index Seek (NonClustered)  
[Inventory].[idx_inv_qty_inge] [inv]  
Cost: 29 %  
0.000s  
32 of  
32 (100%)]
    NL --> CIS[Clustered Index Seek (Clustered)  
[Ingredient].[PK_Ingreديه_B0E453C...]  
Cost: 70 %  
0.000s  
32 of  
32 (100%)]
```

Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour
1	Inventory	R	32	320	960
2	Ingredient	R	32	320	960
Total References			64	640	1,920

4. Identify orders handled by a specific staff member (After: Covering Index)

Transaction Analysis Form					
					April 27, 2026
Transaction	Identify orders handled by a specific staff member (After: Covering Index)				
Volume	Average:	5 per hour			
	Peak:	20 per hour (during report generation periods)			
<pre>CREATE INDEX idx_orders_staffid ON Orders(staff_id) INCLUDE (order_date, order_status, total_price, order_type); GO CREATE PROCEDURE GetOrdersByStaff @staff_id INT AS BEGIN SELECT o.order_id, o.order_date, o.order_status, o.total_price, o.order_type, s.first_name, s.last_name FROM Orders o JOIN Staff s ON o.staff_id = s.staff_id WHERE o.staff_id = @staff_id ORDER BY o.order_date DESC; END; GO --Execute EXEC GetOrdersByStaff @staff_id = 10006; GO</pre>			Predicate:	o.staff_id = @staff_id	
			Join attributes:	Orders.staff_id = Staff.staff_id	
			Ordering attribute:	o.order_date DESC	
			Group attribute:	-	
			Built-in functions:	-	
Attributes updated:	-				

Access	Entity	Type of Access	N0. Of References		
			Per Transaction	Per Transaction	Peak Per Hour

1	Orders	R	726	3,630	14,520
2	Staff	R	1	5	20
Total References			727	3,635	14,540

7. Performance Comparison: Before vs After Optimization

1. Update order status

Metric	Before	After	Improvement / Remark
Table	Orders (25,530 rows)	Orders	—
Physical Operation	Clustered Index Update (PK)	Clustered Index Update + Clustered Index Seek (PK)	Still uses PK seek — fast lookup retained
Rows Read	1	1	No change (already optimal)
Estimated Operator Cost	0.0132842 (100%)	0.0032831 (14%)	~75% cost reduction
Estimated I/O Cost	0.01	0.003125	~69% I/O reduction
Estimated CPU Cost	0.000001	0.0001581	Slight increase (negligible)
Optimization Technique	None (PK already clustered)	Parameterized query using @order_id	Reduced implicit conversion overhead

2.Top-selling menu items

Metric	Before	After	Improvement / Remark
Table	Order_Menu (132,505 rows), Menu (75 rows)	Same	—
Physical Operation (Order_Menu)	Clustered Index Scan	Index Scan (NonClustered) — idx_ordermenu_menuid_qty	Covering index avoids full table scan overhead
Rows Read (Order_Menu)	132,505	132,505	Same — full scan required for GROUP BY
Estimated I/O Cost	0.497199	0.351273	~29% I/O reduction
Estimated Operator Cost	0.643112 (41%)	0.497186 (84%)	~23% operator cost
Aggregation Method	Hash Match (Aggregate)	Stream Aggregate	Stream Aggregate is more efficient (data pre-sorted)
Menu Table Access	Clustered Index Seek (PK)	Clustered Index Seek (PK)	No change — already optimal
Optimization Technique	No covering index	Created index (menu_id, quantity)	Eliminates key lookup

3.Ingredients below minimum stock

Metric	Before	After	Improvement / Remark
Table	Inventory (121 rows), Ingredient (126 rows)	Inventory (32 rows), Ingredient (32 rows)	Reads fewer rows after index
Physical Operation (Inventory)	Clustered Index Scan	Index Seek (NonClustered) — idx_inv_qty_inge	Uses index on quantity_available
Physical Operation (Ingredient)	Clustered Index Scan	Clustered Index Seek	More efficient lookup by ingredient_id
Rows Read	247	64	~74% rows read reduction
Estimated I/O Cost	0.003125	0.003125	No change because dataset is small
Estimated Operator Cost	0.0034151	0.0033172	~3% operator cost reduction
Join Strategy	Merge Join (Inner Join)	Nested Loops (Inner Join)	Better after filtered rows are smaller
Sort Operation	Explicit Sort	Eliminated	Index provides ordered data
Optimization Technique	No index on quantity_available	Created non-clustered index on (quantity_available, ingredient_id)	Reduces scan and removes sort

4. Identify orders handled by a specific staff member

Metric	Before	After	Improvement / Remark
Table	Orders (25,531 rows), Staff (1 row)	Orders (726 rows), Staff (1 row)	Reads fewer Orders rows after index
Physical Operation (Orders)	Clustered Index Scan	Index Seek (NonClustered) — idx_orders_staffid	Uses index on staff_id
Rows Read (Orders)	25,531	726	Much fewer rows read
Rows Returned (Orders)	726	726	Same result output
Estimated I/O Cost	0.15794	0.006342	~96% I/O reduction
Estimated Operator Cost	0.186181 (82%)	0.0072998 (20%)	~96% operator cost reduction
Join Strategy	Nested Loops (Inner Join)	Nested Loops (Inner Join)	No change
Staff Table Access	Clustered Index Seek (PK), 1 row	Clustered Index Seek (PK), 1 row	Already optimal
Sort Operation	Explicit Sort	Explicit Sort	Still needed for ORDER BY o.order_date DESC
Optimization Technique	No index on staff_id	Created covering index on staff_id including order columns	Reduces scan and improves filtering

8. The discussion on the results.

Transaction 1 Update order status

Transaction 1 already used a Clustered Index Seek on `order_id` (Primary Key) before any additional index was created. This confirms that the existing PK clustered index provides efficient single-row lookup — only 1 row is read per execution, with an estimated operator cost of 0.0132842. The query optimizer naturally leverages the PK for the WHERE clause predicate, making a full table scan unnecessary regardless of table size.

After creating an additional non-clustered index on `order_status`, the execution plan for the UPDATE itself remains unchanged (still Clustered Index Seek on PK), which is expected since the WHERE clause filters by `order_id`. However, the additional index on `order_status` further supports other queries in the system that filter or report by order status — for example, listing all Pending or Completed orders — without affecting the performance of T1 itself.

Conclusion: Transaction 1 was already efficient before optimization. The PK index ensures $O(\log n)$ row access. The supplementary index benefits status-based queries elsewhere in the application.

Transaction 2 Top-selling menu items

Before optimization, the query performed a full Clustered Index Scan across all 132,505 rows in `Order_Menu`, followed by a Hash Match Aggregate to compute $\text{SUM}(\text{quantity})$ per `menu_id`. Hash aggregation requires building an in-memory hash table over the entire dataset, which is costly at this data volume.

After creating the covering non-clustered index `idx_ordermenu_menuid_qty` on (`menu_id`, `quantity`), the optimizer switched to an Index Scan on this narrower structure (21 B row size), and replaced Hash Match with a Stream Aggregate — since data now arrives pre-ordered by `menu_id`, the aggregation can be processed as a single pass. Estimated I/O cost on `Order_Menu` dropped from 0.497199 to 0.351273 (~29% reduction), and overall operator cost from 0.643112 to 0.497186. The Menu table lookup via Clustered Index Seek on PK was already optimal and remained unchanged.

Conclusion: Transaction 2 showed the most meaningful improvement. The covering index reduces I/O by narrowing the row width read, and enables the optimizer to choose Stream Aggregate over Hash Match, reducing memory pressure during daily/weekly sales report generation.

Transaction 3 Ingredients below minimum stock

Before optimization, the query performed a full Clustered Index Scan on both Inventory and Ingredient tables, reading 121 and 126 rows respectively, followed by a Merge Join and an explicit Sort operation to support the ORDER BY quantity_available ASC. Since no index existed on quantity_available, the database engine had to scan all rows and sort them, which increased memory usage.

After creating the non-clustered index idx_inv_qty_ing on (quantity_available, ingredient_id) including last_updated, the optimizer switched to an Index Seek on Inventory and a Clustered Index Seek on Ingredient. The condition (quantity_available < 50) is applied to reduce the number of rows processed, making the query more selective. This allows the optimizer to utilize index-based access and significantly improves performance by lowering I/O and memory usage. The join strategy also changed to Nested Loops, which is more efficient for smaller result sets. Additionally, the Sort operation was eliminated because the index already provides data in ascending order. Rows read were reduced from 247 to 64 (~74% reduction), and operator cost slightly decreased from 0.0034151 to 0.0033172.

Conclusion: Transaction 3 shows improvement mainly in reducing unnecessary row scans and eliminating the Sort operation. The index helps optimize ORDER BY and improves efficiency for real-time stock monitoring queries, especially during peak kitchen preparation hours.

Transaction 4 Identify orders handled by a specific staff member

Before optimization, the query performed a Clustered Index Scan on the Orders table, reading 25,531 rows to return only 726 matching rows, along with a Clustered Index Seek on Staff. This resulted in unnecessary data scanning and higher I/O cost.

After creating a non-clustered covering index idx_orders_staffid on staff_id including order_date, order_status, total_price, and order_type, the optimizer switched to an Index Seek on Orders, directly retrieving only the relevant 726 rows. The Staff table access remained a Clustered Index Seek and unchanged. Estimated I/O cost dropped significantly from 0.15794 to 0.006342 (~96% reduction), and operator cost from 0.186181 to 0.0072998.

Conclusion: T4 demonstrates a substantial performance improvement. The covering index enables efficient filtering by staff_id and avoids scanning unnecessary rows, making it highly suitable for frequent reporting queries that retrieve staff-specific orders.

General Observation

Overall, the results demonstrate that index effectiveness depends on both data volume and query selectivity, as well as query structure.

Transaction 1 was already optimal due to the use of a clustered primary key on `order_id`, which enabled efficient single-row lookup using a Clustered Index Seek. The additional non-clustered index on `order_status` did not improve this transaction directly, but provides benefits for other queries filtering by order status.

Transaction 2 showed a clear improvement in I/O efficiency. Although the number of rows read remained the same, the covering non-clustered index on (`menu_id`, `quantity`) reduced row width and allowed the optimizer to switch from Hash Match to Stream Aggregate. This eliminated the need for memory-intensive hash operations and improved aggregation efficiency on large datasets.

Transaction 3 demonstrated performance improvement through a combination of index optimization and query refinement. The introduction of a non-clustered index on `quantity_available` reduced row scanning and eliminated the explicit Sort operation. In addition, the inclusion of a filtering condition (`WHERE quantity_available < 50`) significantly reduced the number of rows processed. Therefore, the improvement is attributed to both indexing and query rewriting.

Transaction 4 showed the most significant reduction in I/O cost. Before optimization, the query performed a full table scan on `Orders`, reading all rows before filtering. After introducing a covering index on `staff_id`, the optimizer used an Index Seek to directly retrieve only the required rows. This eliminated unnecessary scanning and resulted in substantial performance gains.

These results highlight that the SQL Server query optimizer is cost-based and selects execution strategies based on estimated row counts, I/O cost, and query structure. Indexes are most effective when they align with query predicates and reduce unnecessary data access, particularly in high-volume tables and selective queries.